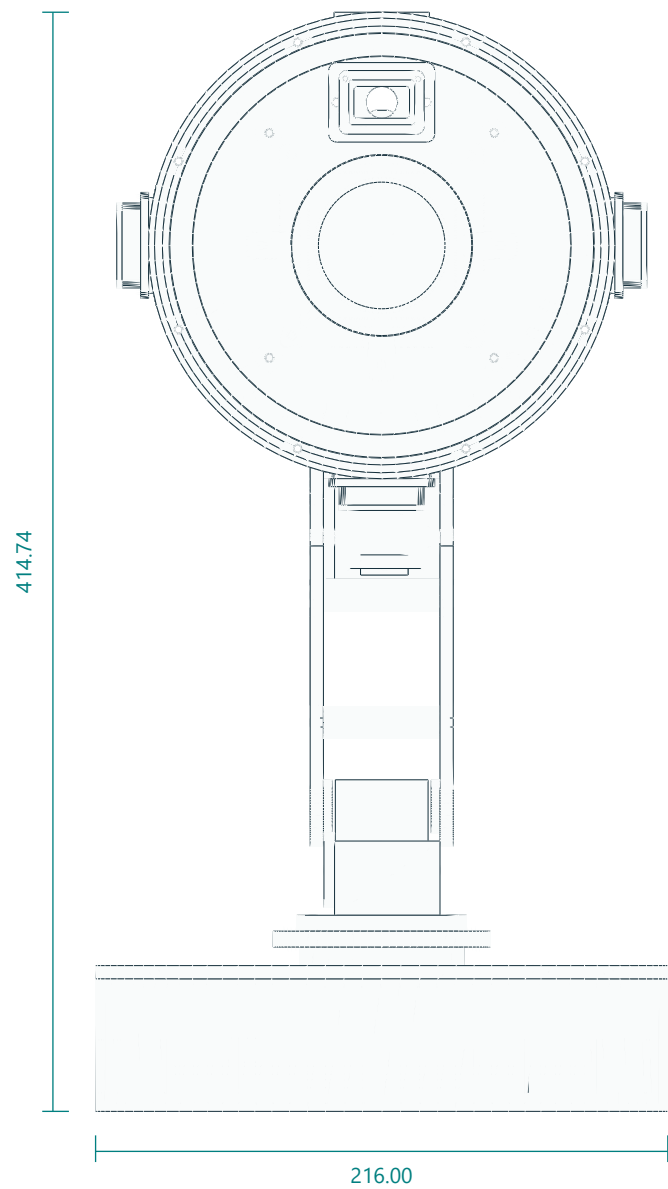


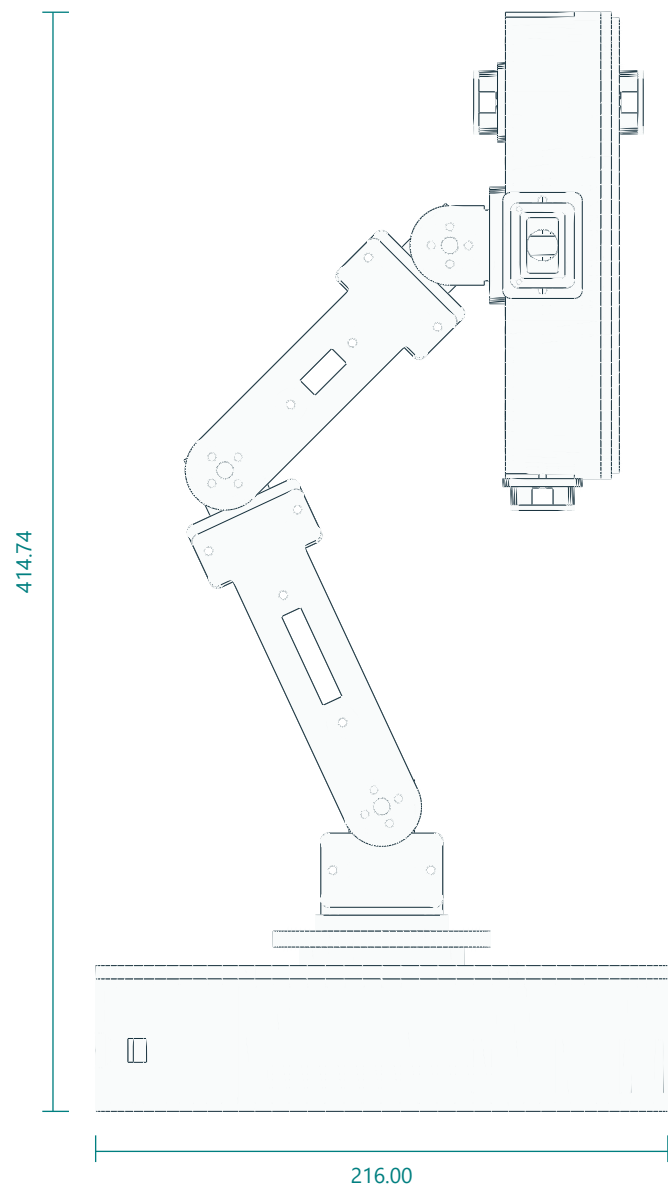
Finished assembly / indexed neutral pose

Assembled geometry from the same STL files supplied for printing. Purchased parts are simplified envelopes.

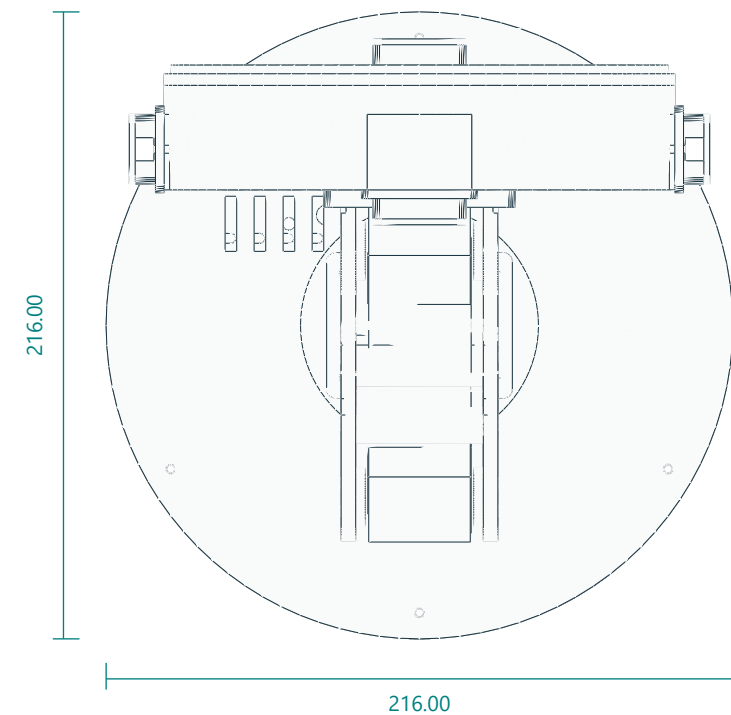
FRONT / LCD FACING VIEW



RIGHT SIDE / NEUTRAL POSE



TOP / FOOTPRINT

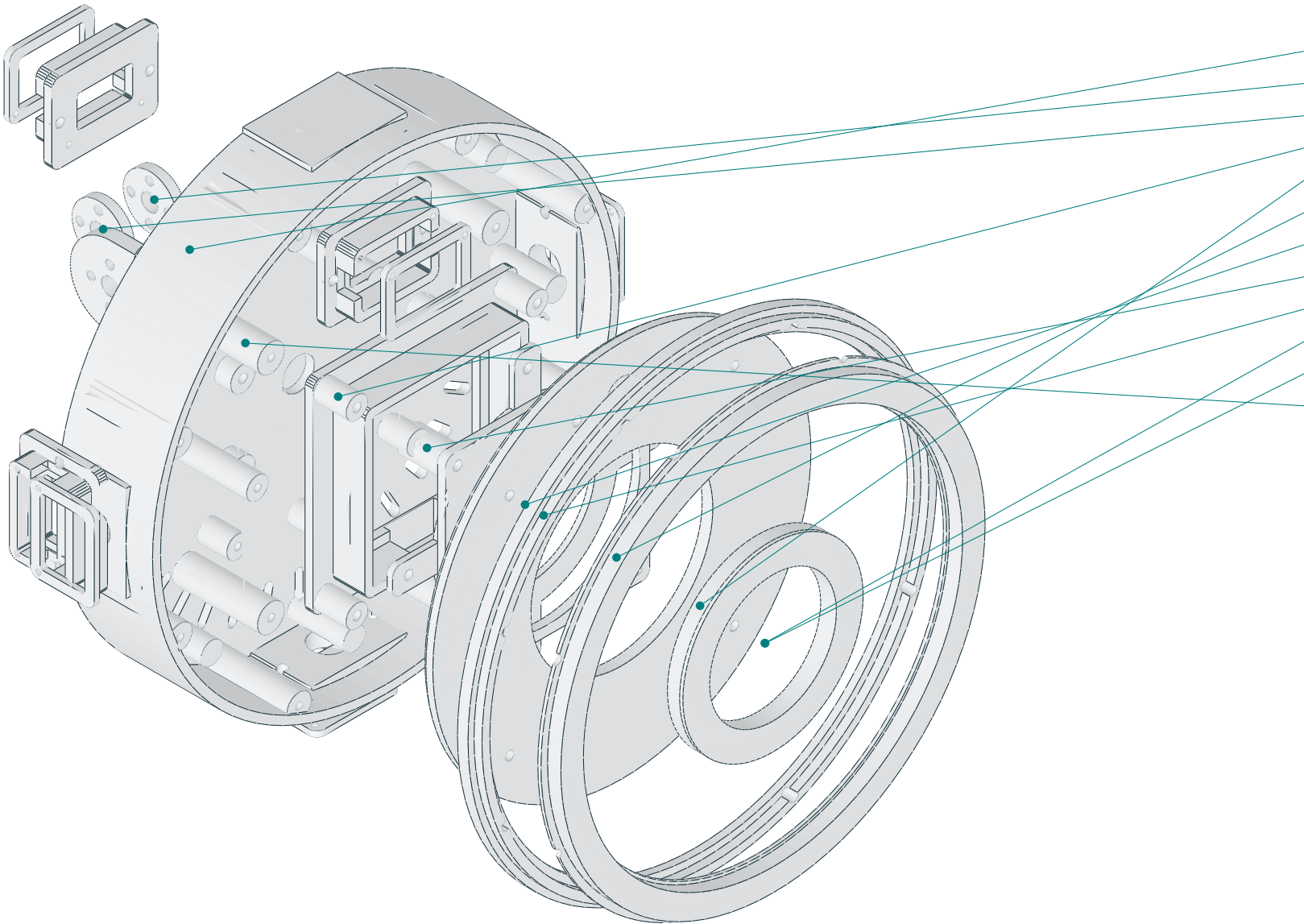


Four axes: yaw about Z; shoulder, elbow and wrist about local X. Datum is the base underside at Z=0.
Nominal linkage pitches: shoulder–elbow 140 mm; elbow–wrist 120 mm. The illustrated height is a pose dimension, not a motion envelope.
Neutral: shoulder link 115° above horizontal; forearm 45° above horizontal; face directed toward +Y. Follow indexed calibration in the manual.
Do not infer screw lengths from the illustration. Fasteners and service loops are specified in the assembly chapter.

Head / layered assembly study

Exploded positions expose the optical parts; displacements are illustrative and are not insertion paths.

EXPLODED HEAD / MODEL-DERIVED VIEW



LAYER REFERENCES

- 01 Head tilt fork
- 02 head left horn spacer
- 03 head right horn spacer
- 04 Head Shell
- 05 Outer Bezel
- 06 Face Center
- 07 White Carrier
- 08 Lcd Cradle
- 09 Lcd Retainer
- 10 Outer Diffuser
- 11 Inner Diffuser
- 12 Purchased LCD envelope

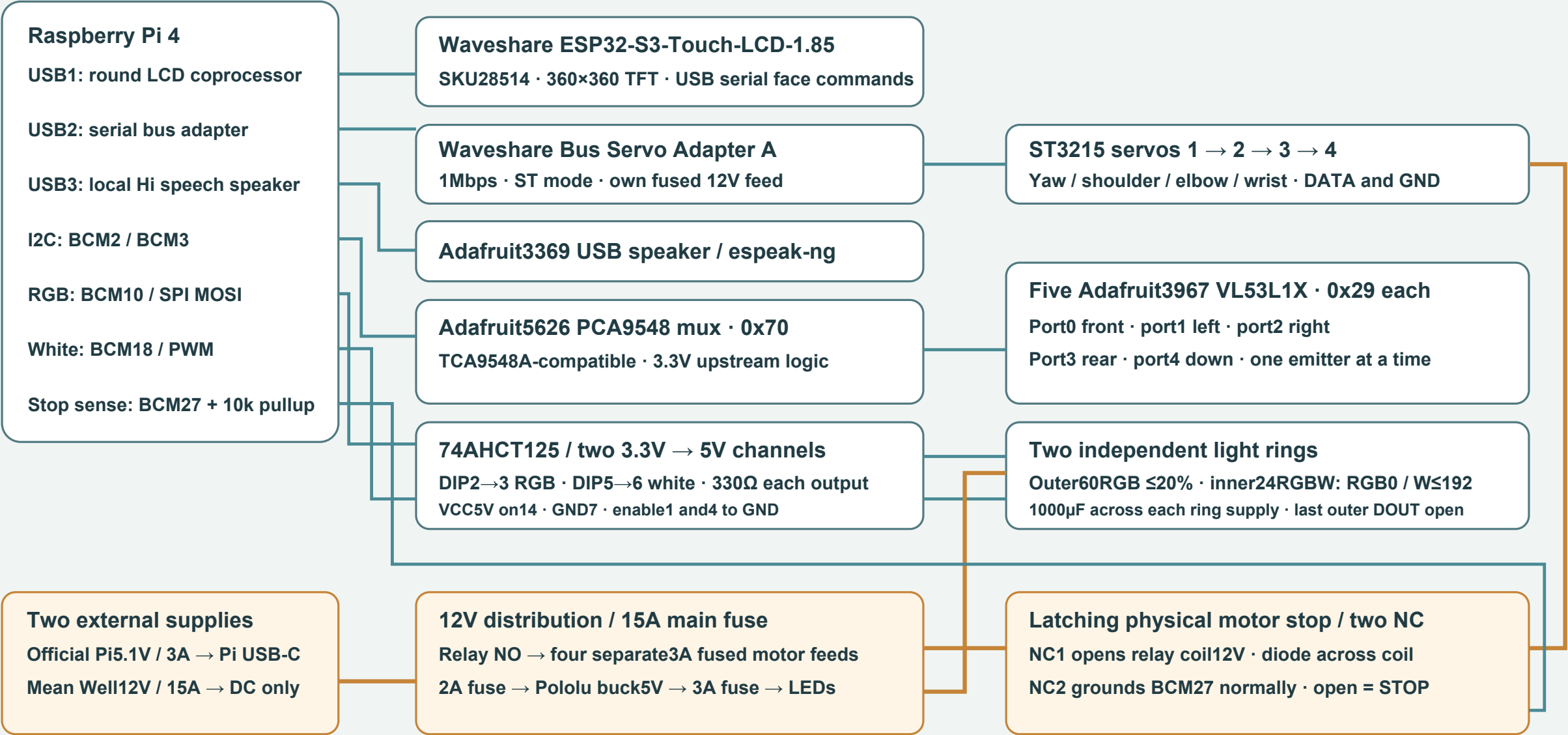
Central LCD board is behind the inner white ring. Keep all optical components at their documented axial heights.
Use natural PETG for both light diffusers and opaque PETG for the optical baffle. Leave LCD glass and all ToF apertures clear.
The four outer ring PCB quarters require continuous mechanical support. Solder joints are electrical connections only.
Refer to the head assembly chapter for screw order, spacers, wire relief and board retention. Purchased LEDs are not printable parts.

Electrical connections / system wiring

Follow the point-to-point wiring schedule for connector pins, polarity and conductor ratings.

LUMA / Electrical connections

Revision A · Raspberry Pi 4 master · four bus servos · exactly five ranging sensors



Read wiring.csv for exact connections, fuses, DIP pins and cable notes.

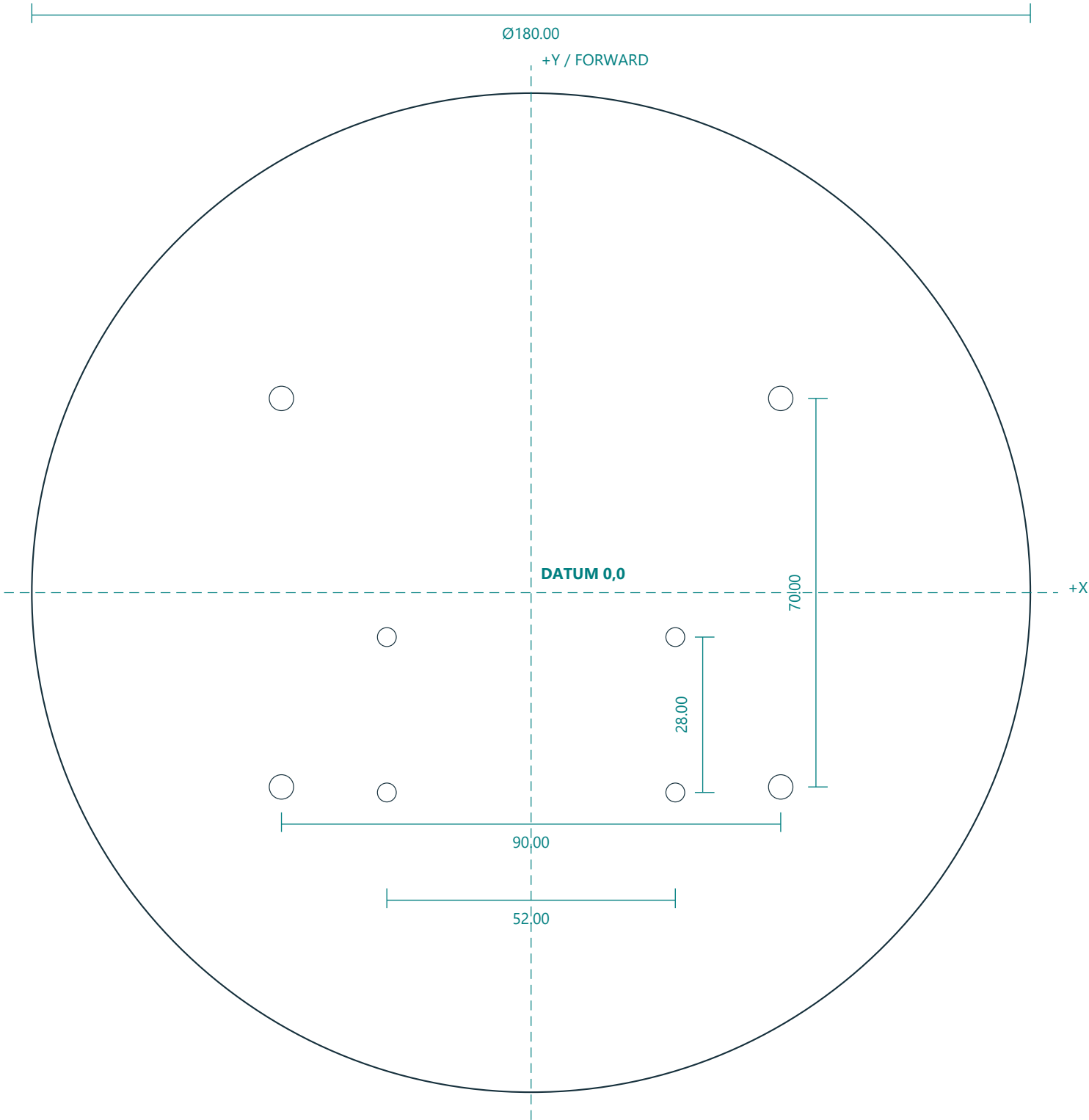
All DC grounds meet at one star point. Never join Pi5V and LED5V. Never carry all motor current through the adapter jack.

Motor stop cuts torque: support the arm. Five narrow ToF cones are interaction inputs and do not cover every pinch point.

Power wires shown in amber; signal routes in blue. Schematic overview: routing lengths and connector positions are not to scale.

Steel ballast plate / fabrication drawing

Purchased/custom-fabricated metal part M01. This sheet is not a printable-plastic substitute.



MATERIAL + INTERFACES

Material: mild steel; finished thickness 6.00mm .
Outer diameter 180.00mm ; deburr all edges and holes.
Datum origin is the disc center. All hole axes normal to plate.
 $4 \times \varnothing 4.40$ through: $X = \pm 45.00$, $Y = \pm 35.00$.
 $4 \times \varnothing 3.40$ through: $X = \pm 26.00$, $Y = -36.00$ and -8.00 .
Do not add a center hole. Rear yaw bolt pattern points-Y.
Nominal mass before drilling: 1.19kg at steel density 7.85g/cm^3 .
Fit to printed base floor before final coating. Inspect burrs and screw-head clearance.
Suggested workshop positional allowance $\pm 0.20\text{mm}$; confirm actual printed mating pattern before drilling.

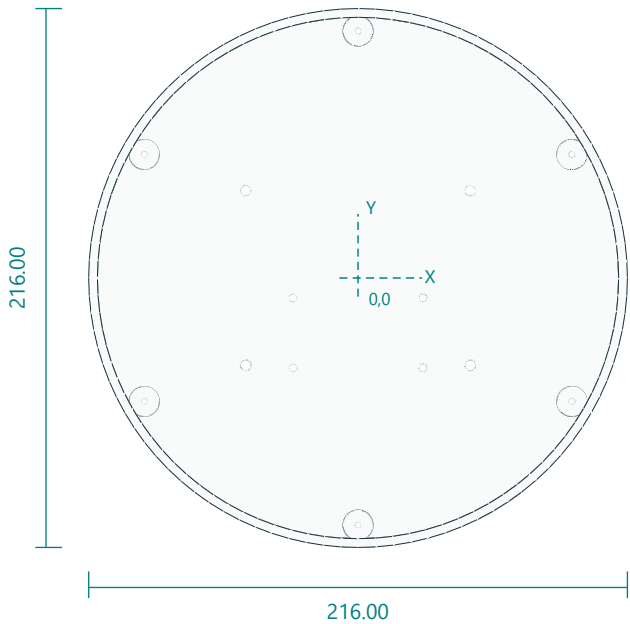
SIDE VIEW / THICKNESS



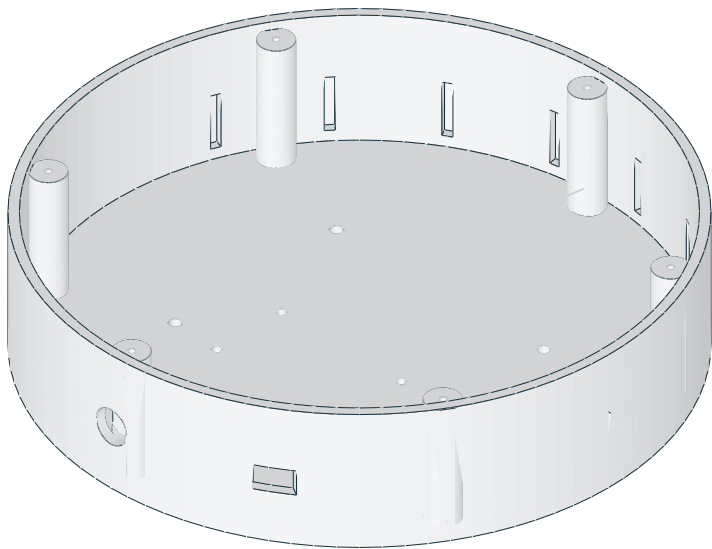
Weighted base enclosure

base_tub.stl | PRINT QUANTITY 1 | PETG

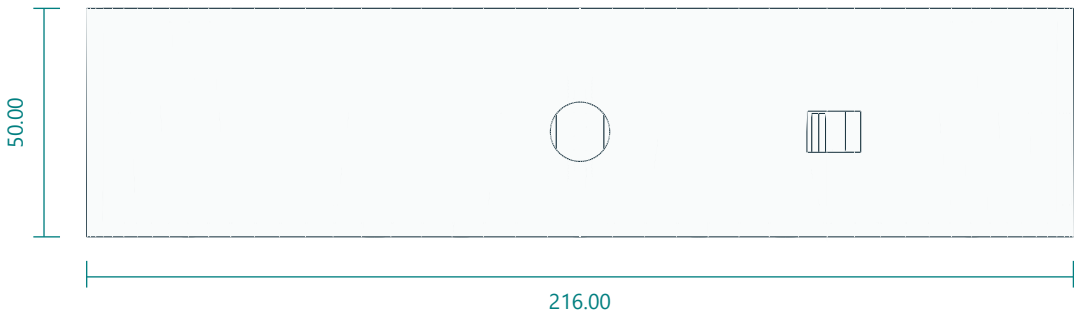
TOP / XY



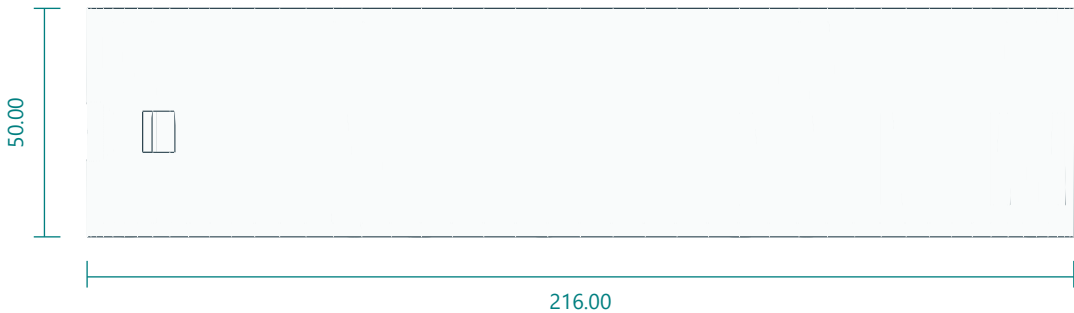
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

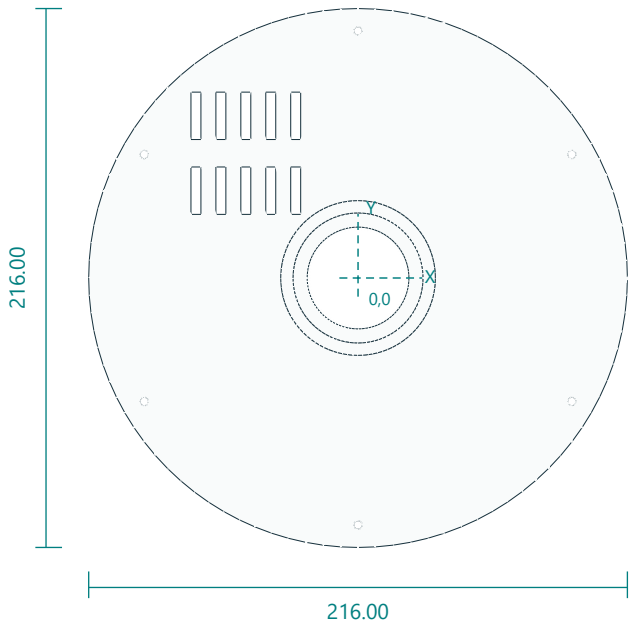


Interfaces: 216 diameter; 50 high; 3 floor; 3.5 wall.
Fasteners / retention: 6 M3x12 lid screws; 4 M4 ballast retainers.
Print orientation: floor down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: M4 clearance4.4 at (+/-45,+/-35). Yaw bolts3.4 at (+/-26,-36) and (+/-26,-8). Six lid pilots2.8 at radius99 angles30+60n deg.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

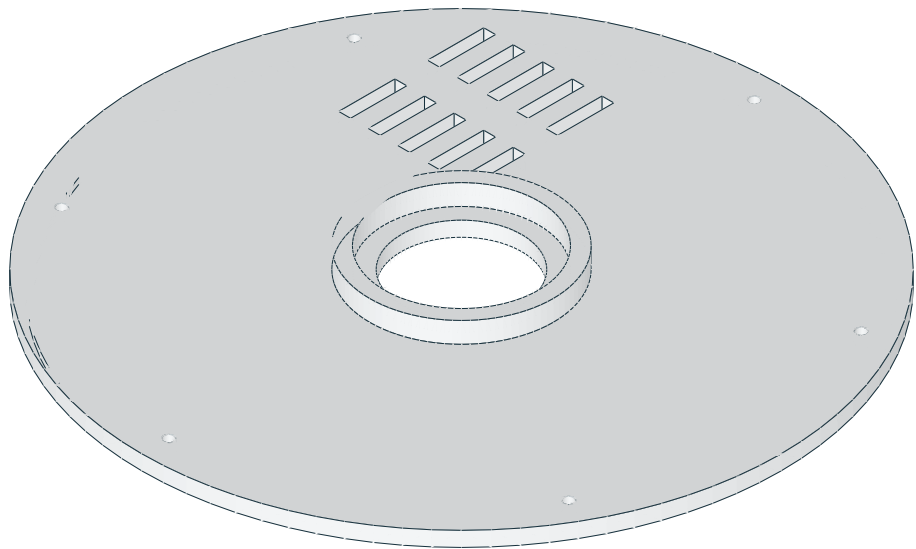
Vented base lid and yaw bearing seat

base_lid.stl | PRINT QUANTITY 1 | PETG

TOP / XY



ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

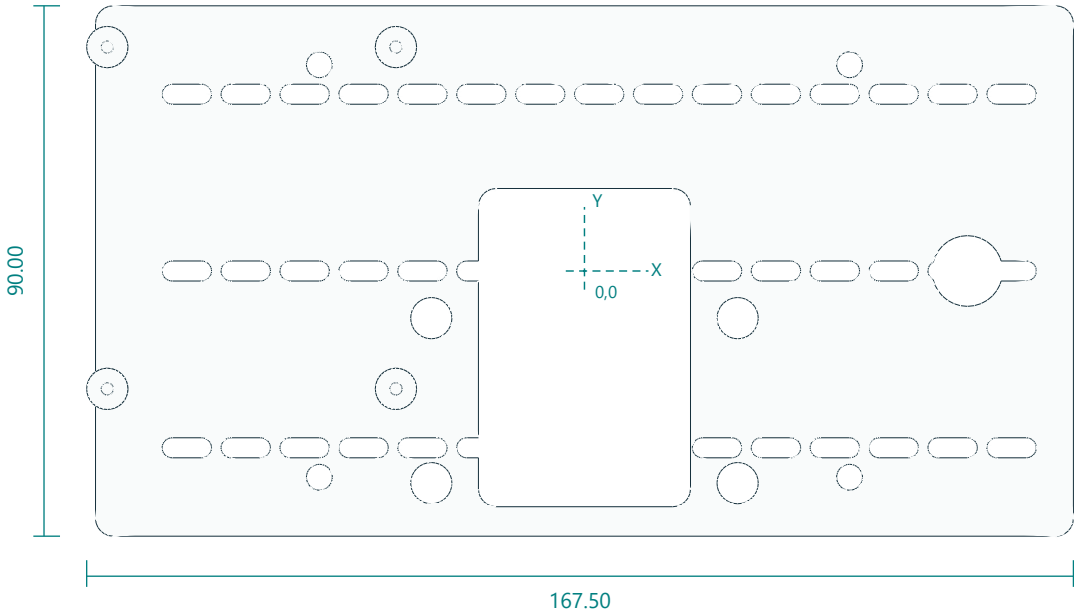


Interfaces: 216 diameter; bearing pocket 52.15 x 7; 40.8 cable/hub bore.
Fasteners / retention: 6 M3x12.
Print orientation: flat annulus down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Six 3.4 holes radius 99 angles 30+60n deg. Bearing 52.15 bore by 7 depth. STL bottom datum is 2mm below principal lid underside.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

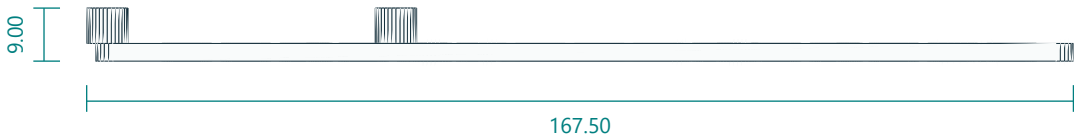
Universal electronics tray with Pi4 standoffs

electronics_tray.stl | PRINT QUANTITY 1 | PETG

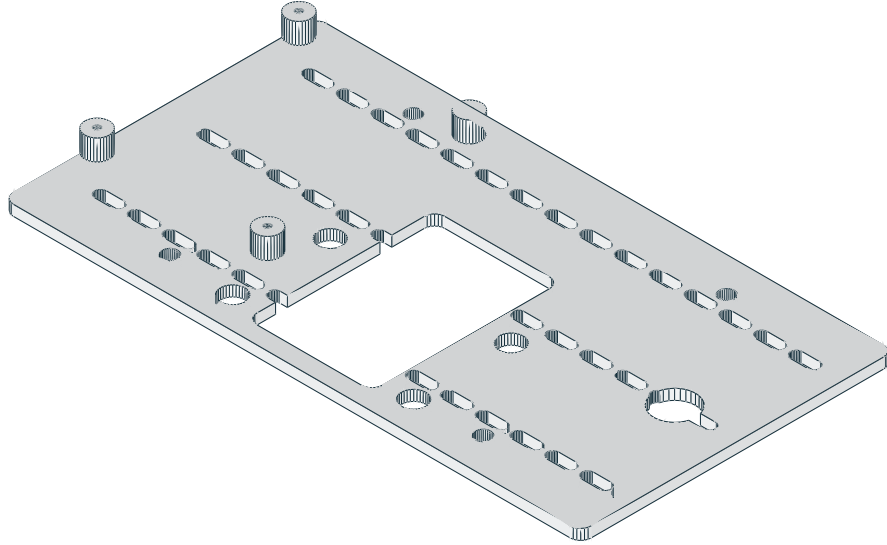
TOP / XY



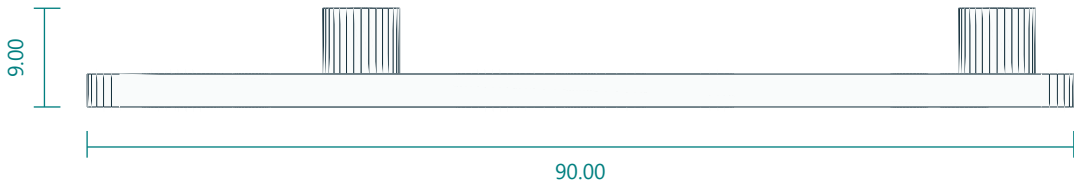
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

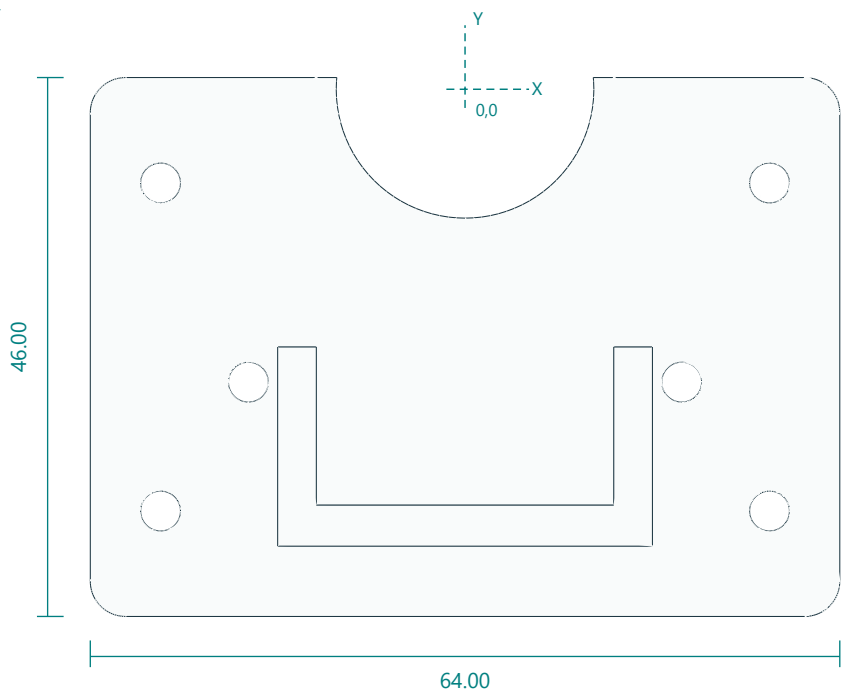


Interfaces: 166 x90; Pi4 hole pitch49 x58; 2.1mm M2.5 tapping pilots.
Fasteners / retention: 4 M2.5x6 Pi; 4 M4x25 with metal spacers.
Print orientation: plate down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: M4 holes (+/-45,+/-35). Pi M2.5 pilots2.1 at X=-81,-32 and Y=-20,38. Central rounded36x54 clearance centered(0,-13). Four7mm tool apertures at yaw bolts.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

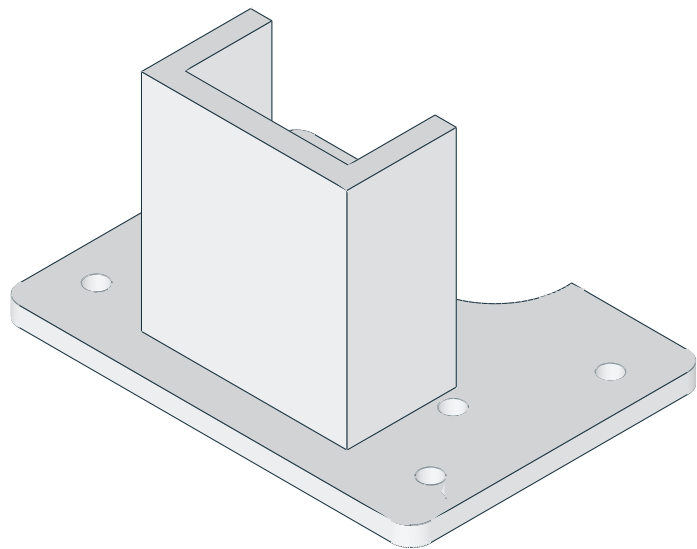
Yaw servo rear body cradle

yaw_mount.stl | PRINT QUANTITY 1 | PETG

TOP / XY



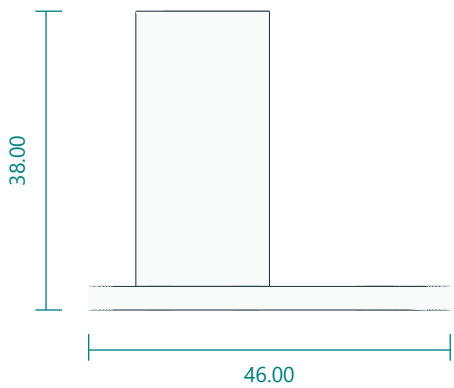
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

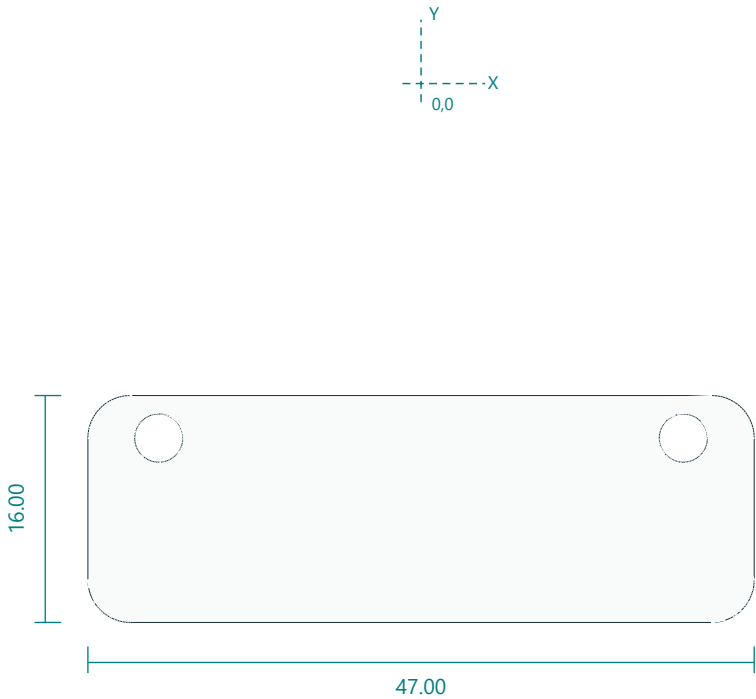


Interfaces: 64 x46; pocket25.4; rear body clamp.
Fasteners / retention: 4 M3x12 mount; 2 M3x45 clamp.
Print orientation: base down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 floor holes (+/-26,-36/-8). Keeper bolts at (+/-18.5,-25). Servo shaft atXY(0,0).
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

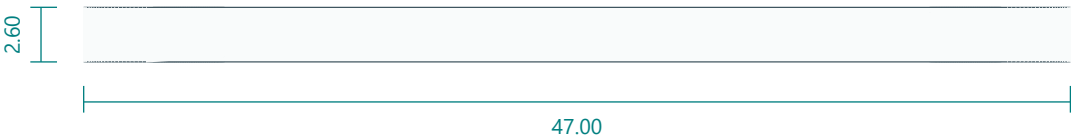
Yaw clamp keeper

yaw_cap.stl | PRINT QUANTITY 1 | PETG

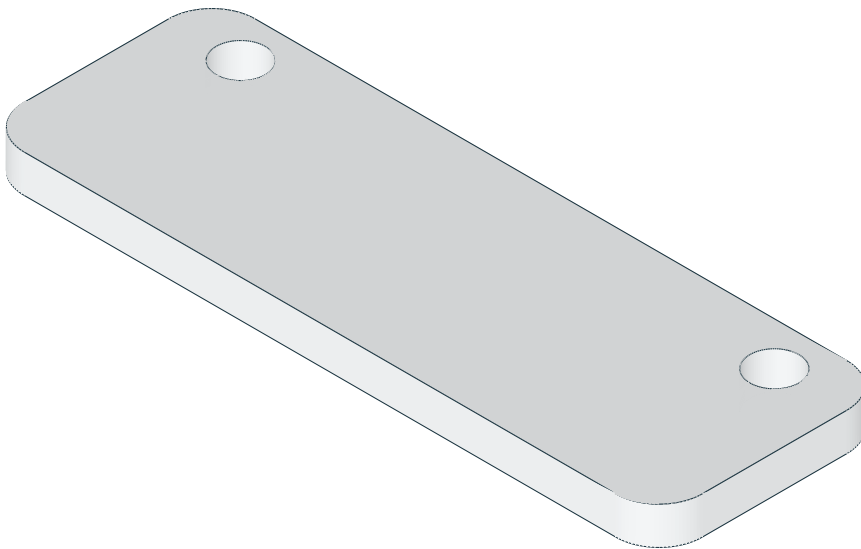
TOP / XY



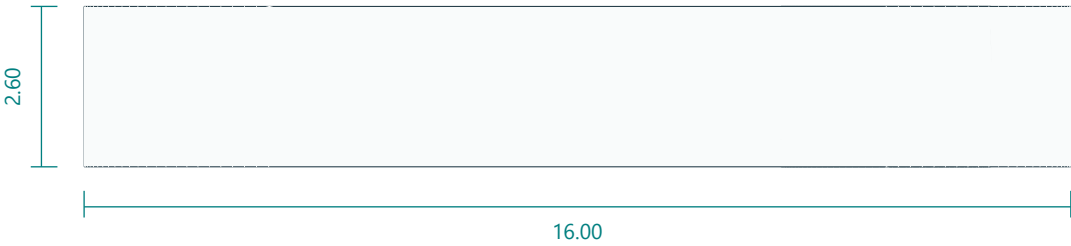
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

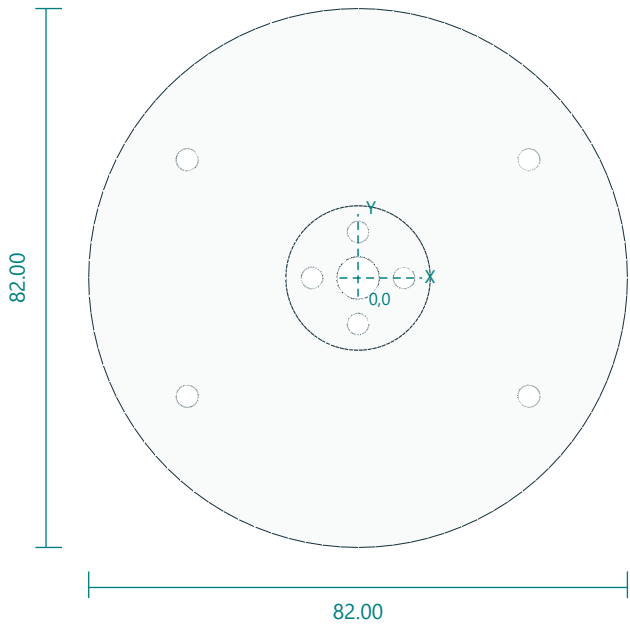


Interfaces: 47 x16 x2.6; clamp hole pitch37.
Fasteners / retention: 2 M3x45.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Two3.4 holes (+/-18.5,-25). Keeper center(0,-30); thickness2.6.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

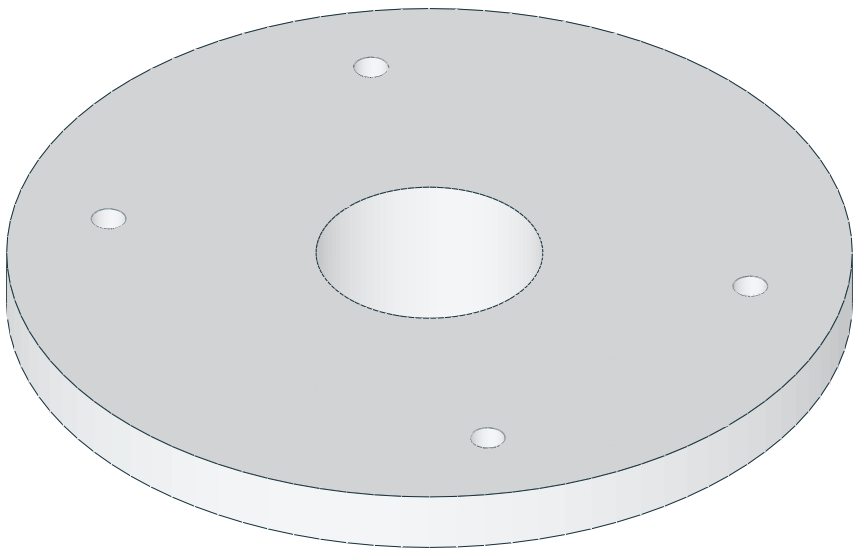
Bearing supported yaw turntable

turntable.stl | PRINT QUANTITY 1 | PETG

TOP / XY



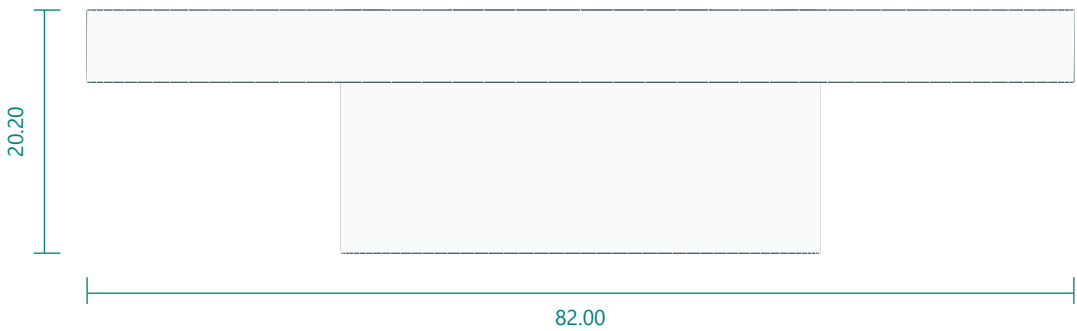
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

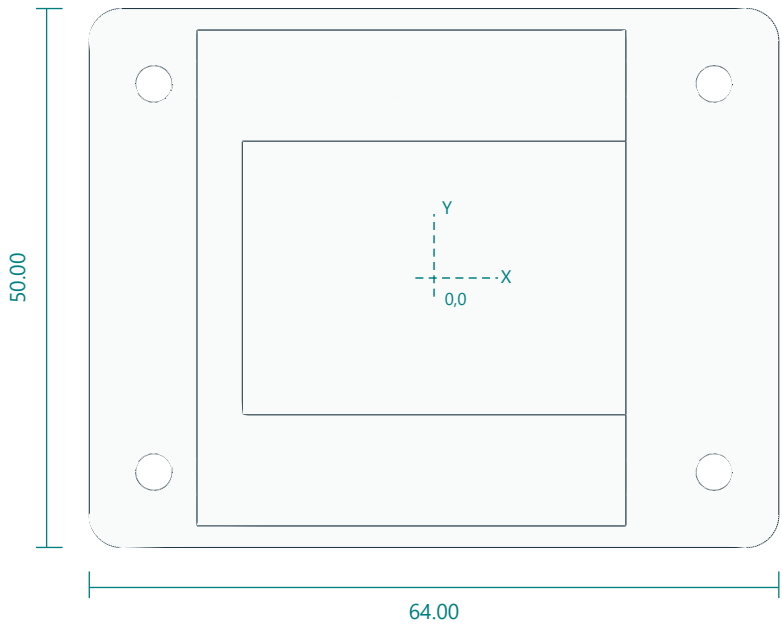


Interfaces: 82 diameter; neck39.75 x14.2; horn PCD14; tower52 x36.
Fasteners / retention: 4 nominal M3x6 horn screws; 4 M3x16 tower.
Print orientation: neck down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Horn4x3.3 at (0,+/-7),(+/-7,0); center6.5 in3mm web. Tower4x3.4 at(+/-26,+/-18). Top central tool aperture22.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

Shoulder servo body tower

shoulder_tower.stl | PRINT QUANTITY 1 | PETG

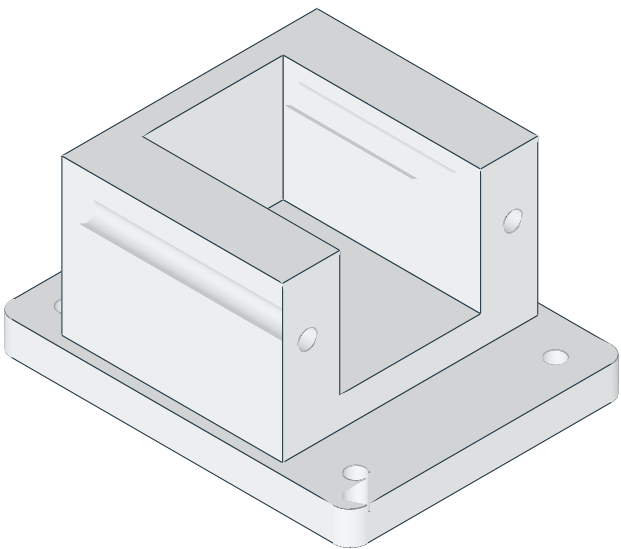
TOP / XY



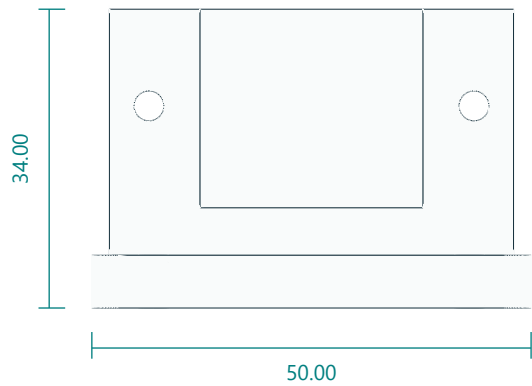
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

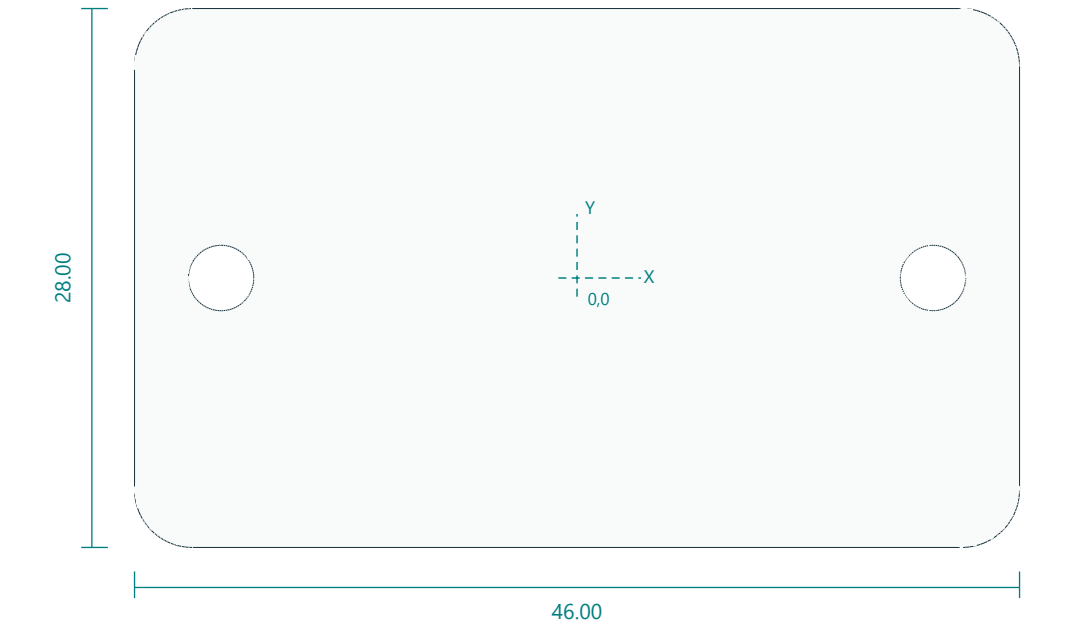


Interfaces: 64 x50; pitch axis47 above foot; body35.6 x25.4.
Fasteners / retention: 4 M3x16; 2 M3x50 clamp.
Print orientation: foot down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Foot4x3.4 at(+/-26,+/-18). Clamp holes alongX atY+/-18.5,Z23. Pitch axis47 abovefoot.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

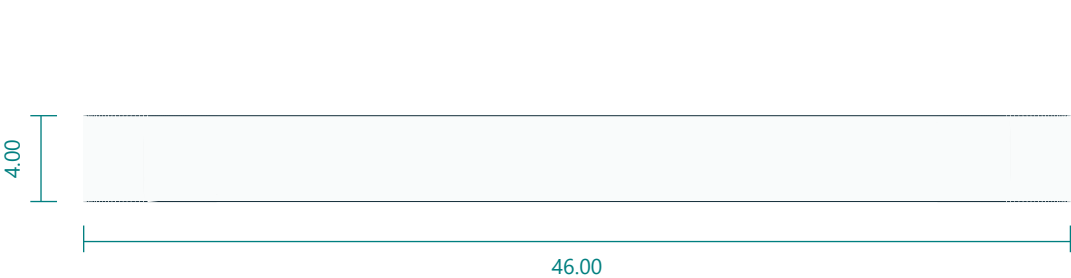
Shoulder clamp keeper

shoulder_cap.stl | PRINT QUANTITY 1 | PETG

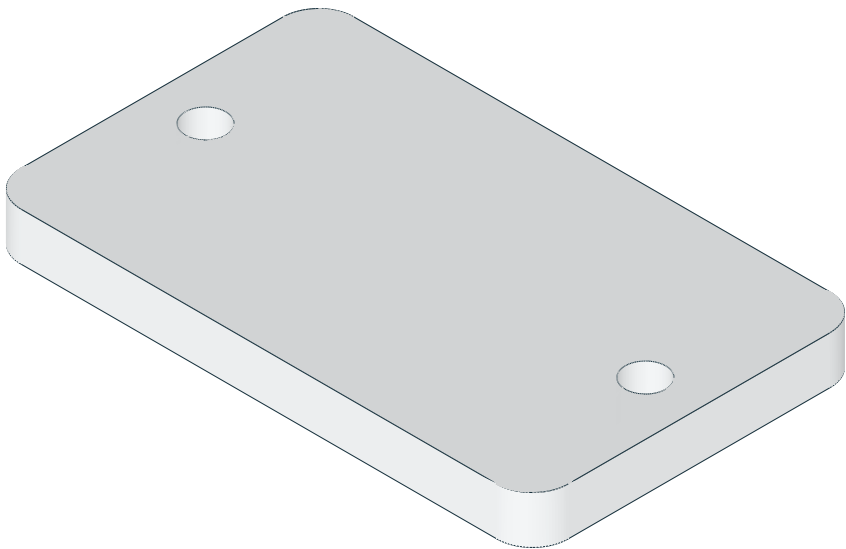
TOP / XY



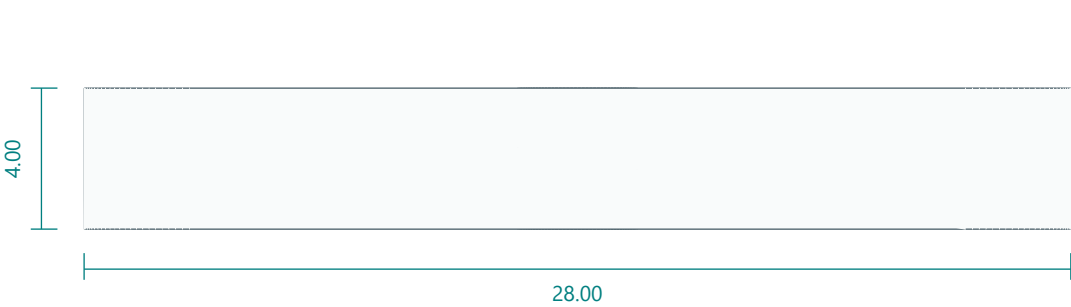
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

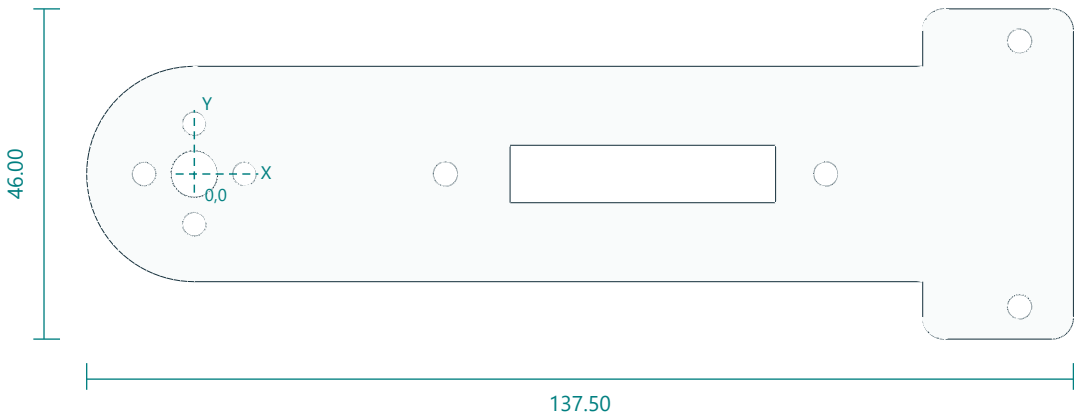


Interfaces: 46 x28 x4; clamp pitch37.
Fasteners / retention: 2 M3x50.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Two3.4 holes (+/-18.5,0). Printedface46x28.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

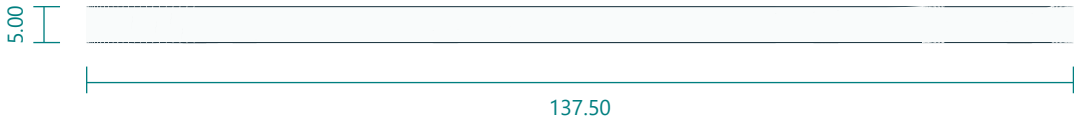
140 mm upper arm side rail

upper_rail.stl | PRINT QUANTITY 2 | PETG

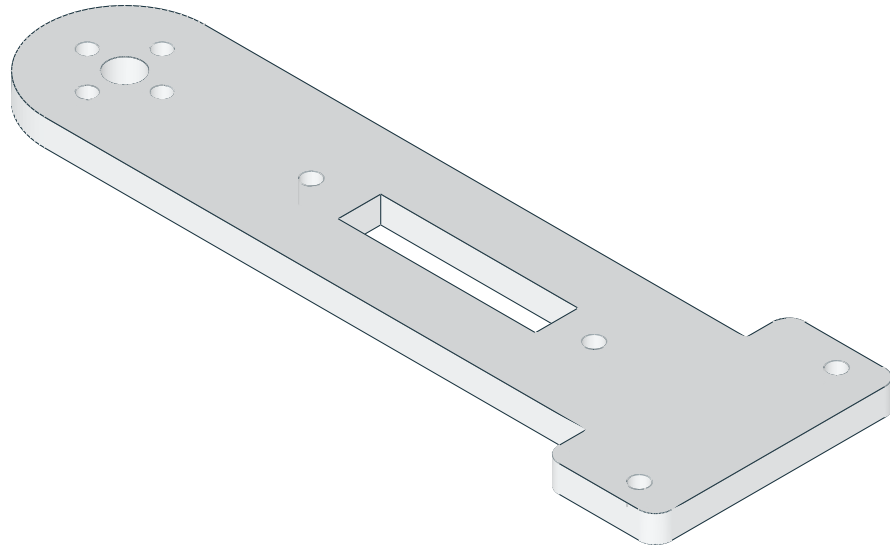
TOP / XY



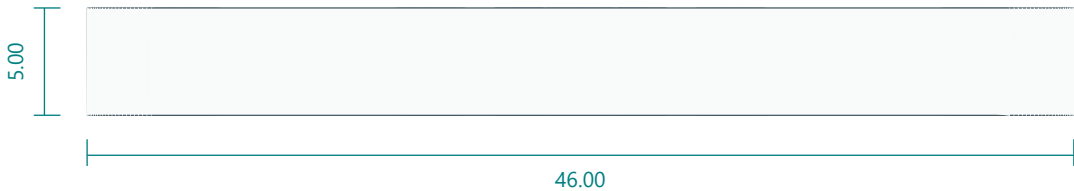
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ



Interfaces: 140 pivot pitch; 5 thick; 14 horn PCD; 3.3 horn holes.
Fasteners / retention: 8 nominal M3x10 horn; 4 M3x60 cross bolts.
Print orientation: broad face down. Layers 0.2 mm; 6 walls; 40% infill baseline.
Feature locations: Proximal horn4x3.3 at(0,+/-7),(+/-7,0). Cross bolts3.4 at(35,0),(88,0). Cassette bolts at(115,+/-18.5). Distal shaft140 from proximal.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

120 mm forearm side rail

forearm_rail.stl | PRINT QUANTITY 2 | PETG

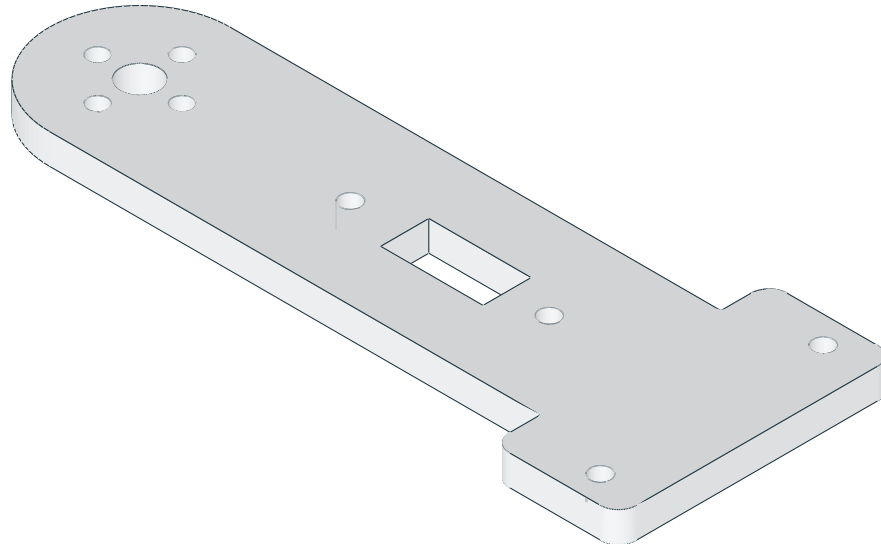
TOP / XY



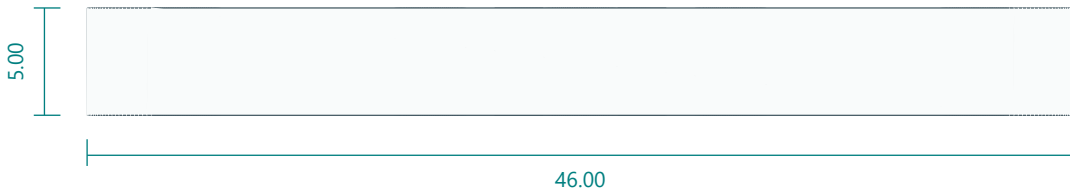
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

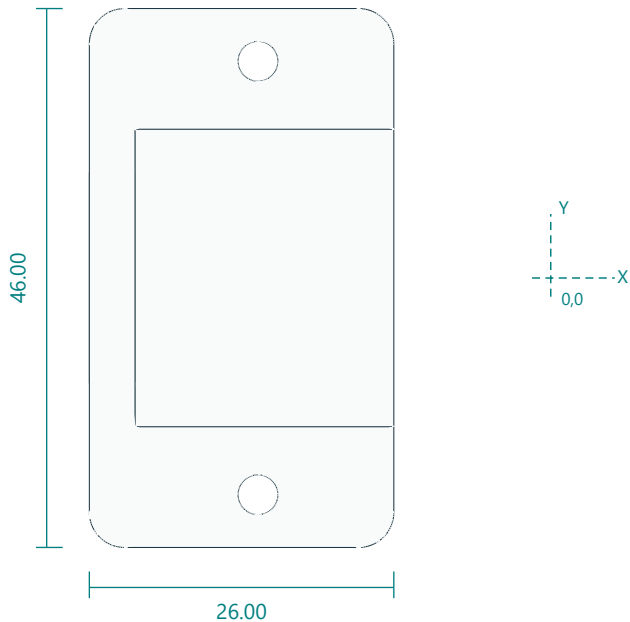


Interfaces: 120 pivot pitch; 5 thick; 14 horn PCD; 3.3 horn holes.
Fasteners / retention: 8 nominal M3x10 horn; 4 M3x60 cross bolts.
Print orientation: broad face down. Layers 0.2 mm; 6 walls; 40% infill baseline.
Feature locations: Proximal horn4x3.3 at(0,+/-7),(+/-7,0). Cross bolts3.4 at(35,0),(68,0). Cassette bolts at(95,+/-18.5). Distal shaft120 from proximal.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

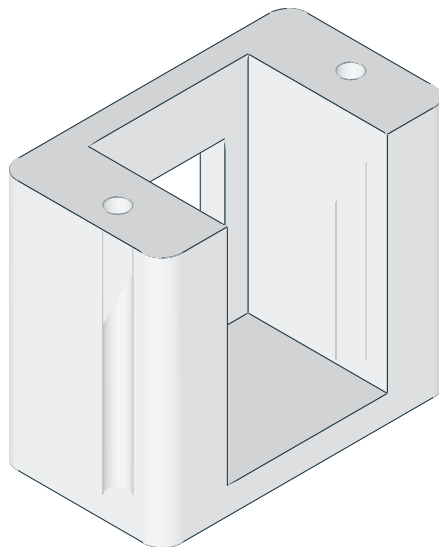
Elbow/wrist rear servo body cassette

servo_cassette.stl | PRINT QUANTITY 2 | PETG

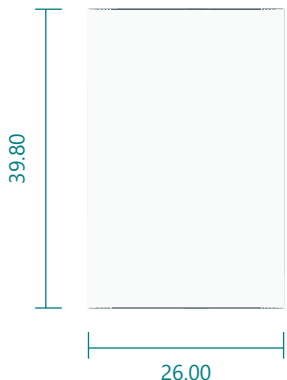
TOP / XY



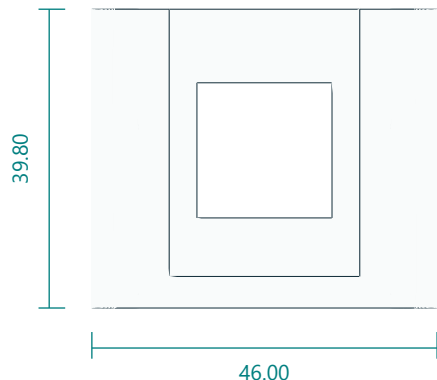
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

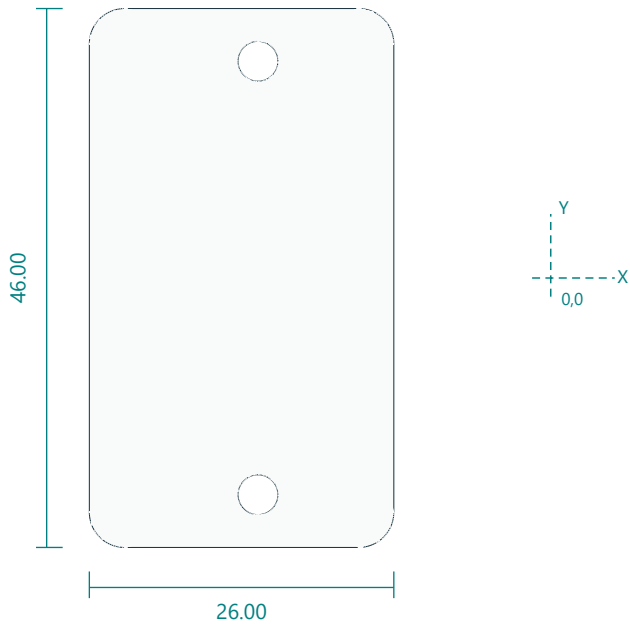


Interfaces: 26 x46 x39.8; 25.4 wide; 35.6 clamped axis width.
Fasteners / retention: 2 M3x60 each shared with rails.
Print orientation: closed side down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Bolts3.4 at(-25,+/-18.5). Body pocketY+/-12.7; floor4.2; assembledcaseaxiswidth35.6.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

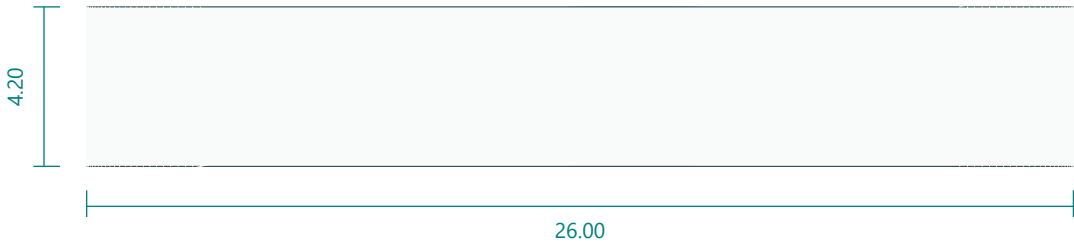
Servo body cassette keeper

cassette_cap.stl | PRINT QUANTITY 2 | PETG

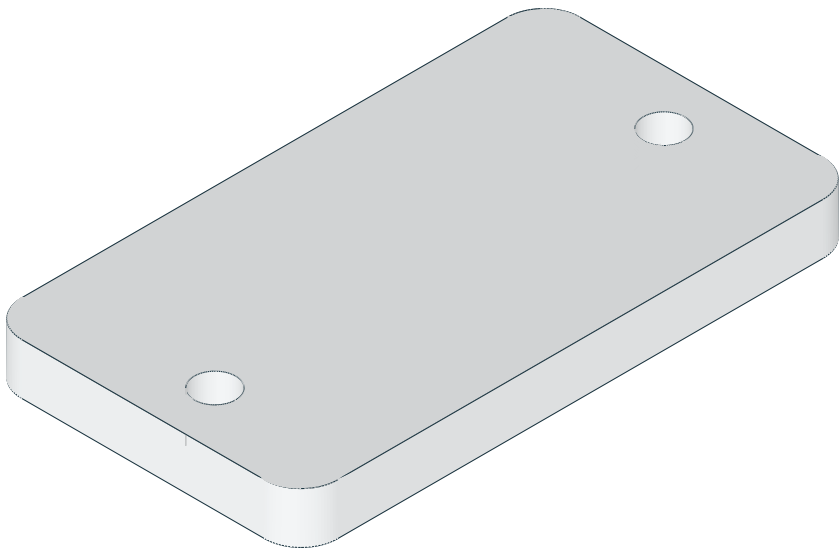
TOP / XY



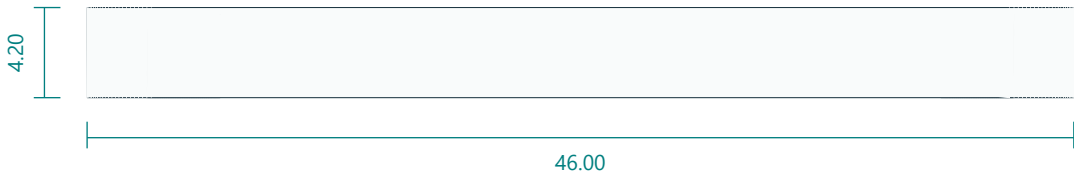
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

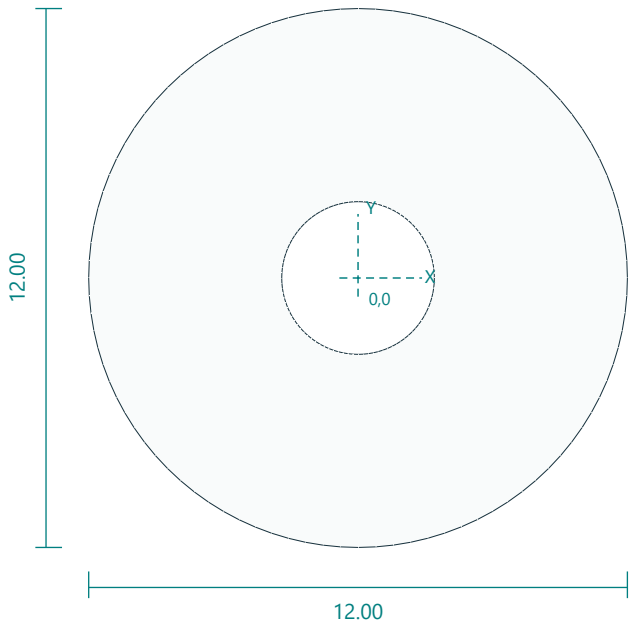


Interfaces: 26 x46 x4.2.
Fasteners / retention: shared M3x60.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Bolts3.4 at(-25,+/-18.5); axialthickness4.2.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

Arm through bolt spacer

cross_spacer.stl | PRINT QUANTITY 4 | PETG

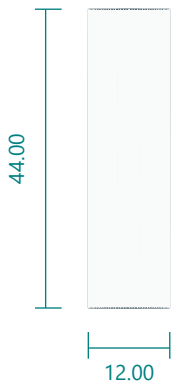
TOP / XY



ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

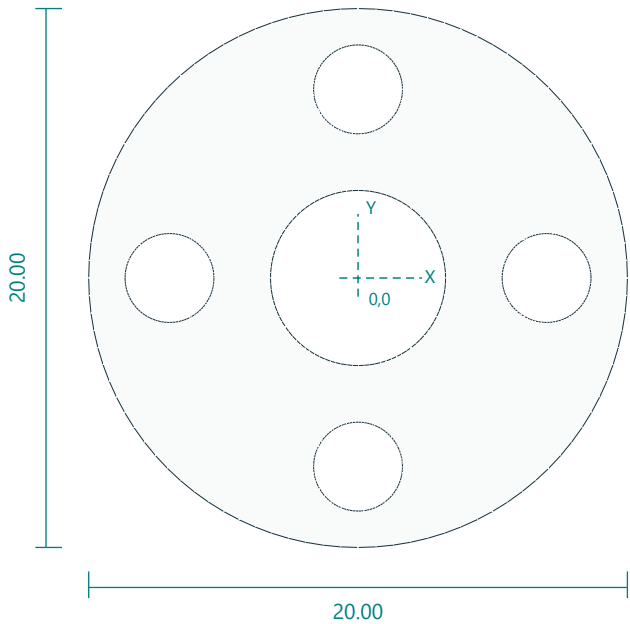


Interfaces: 12 diameter x44; bore3.4.
Fasteners / retention: 1 M3x60 each.
Print orientation: cylinder upright. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Axialbore3.4; length44; OD12.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

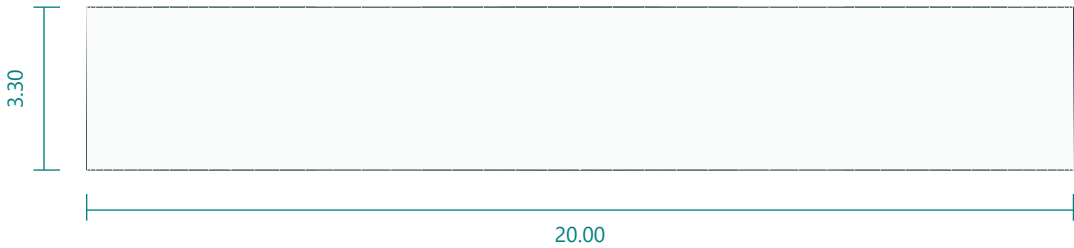
Metal horn to rail stand off

horn_spacer.stl | PRINT QUANTITY 6 | PETG

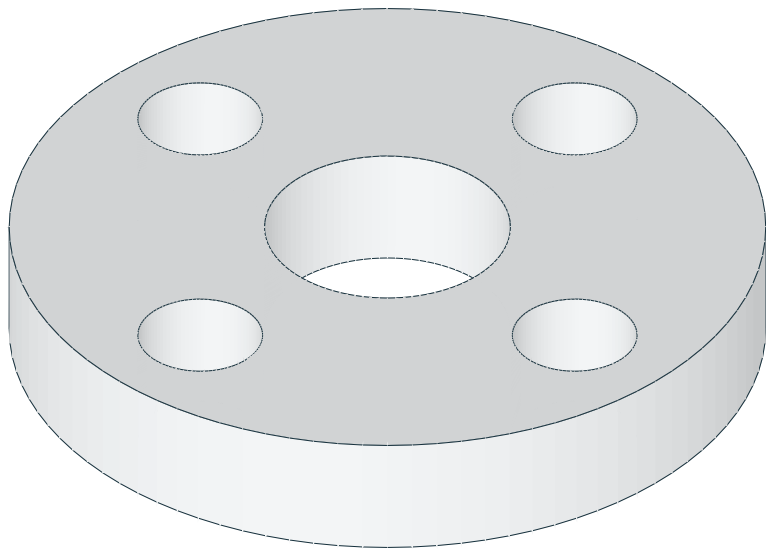
TOP / XY



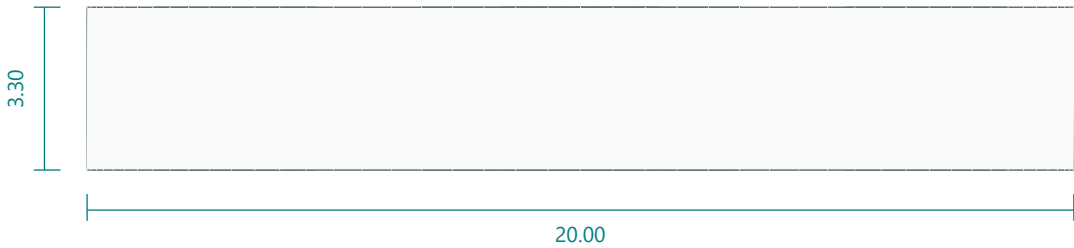
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

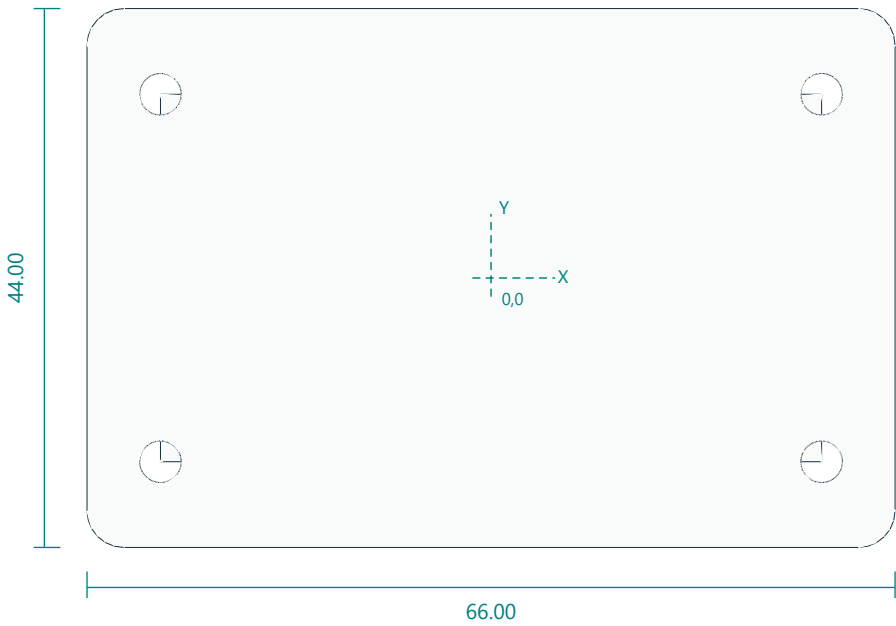


Interfaces: 20 diameter x3.3; PCD14; center6.5.
Fasteners / retention: shared nominal M3x10 horn screws.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: PCD14 four3.3 holes; center6.5; thickness3.3.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

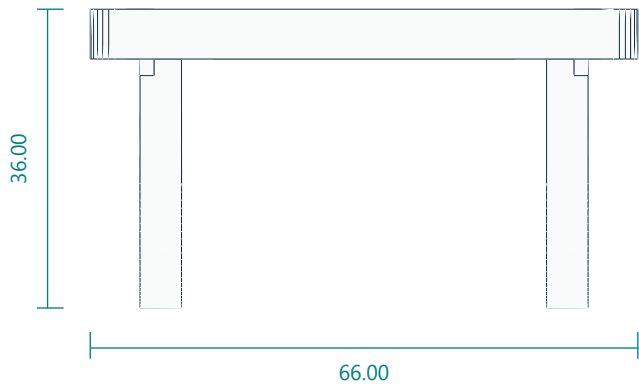
Head tilt fork and rear mount

head_yoke.stl | PRINT QUANTITY 1 | PETG

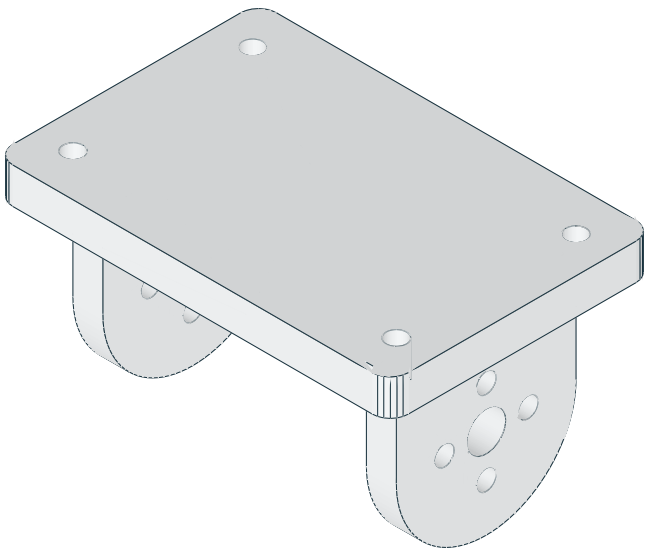
TOP / XY



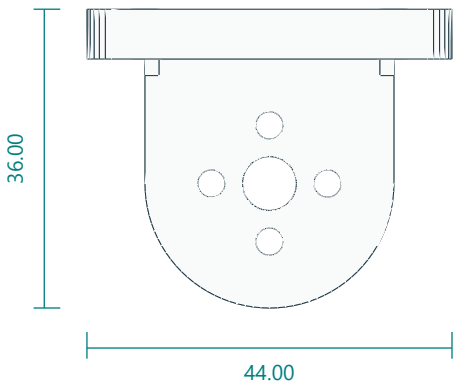
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

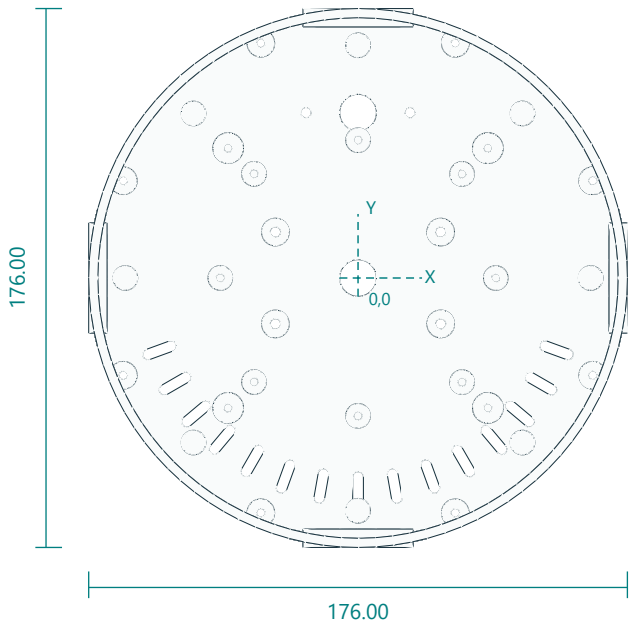


Interfaces: 54 fork span; axis21 behind head rear; rear holes54 x30.
Fasteners / retention: 8 nominal M3 horn screws; 4 M3x12 rear screws.
Print orientation: rear mounting plate on bed; rotate180X. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Back bolts3.4 XY(+/-27, +/-15); attachmentplaneZ36. HornaxesX atY0,Z15, PCD14. Forkinsidewidth44.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

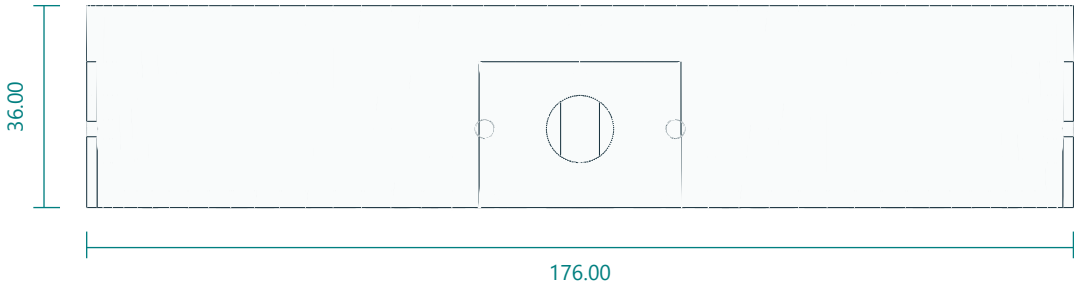
Optical head rear enclosure

head_shell.stl | PRINT QUANTITY 1 | PETG

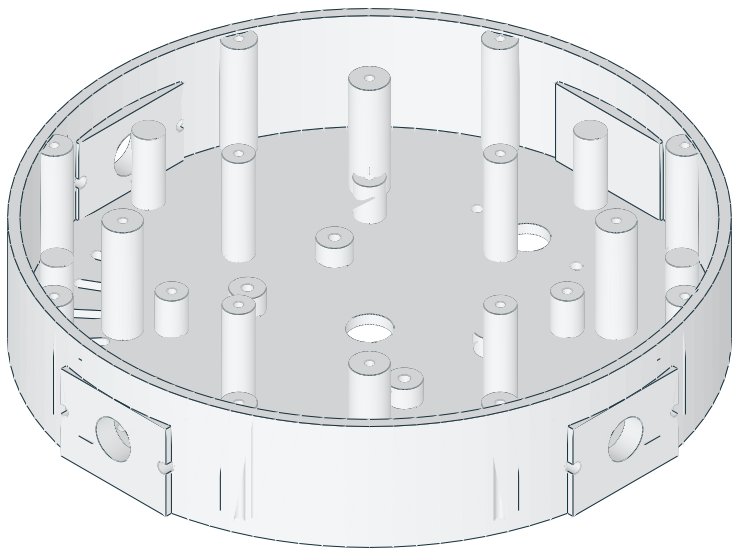
TOP / XY



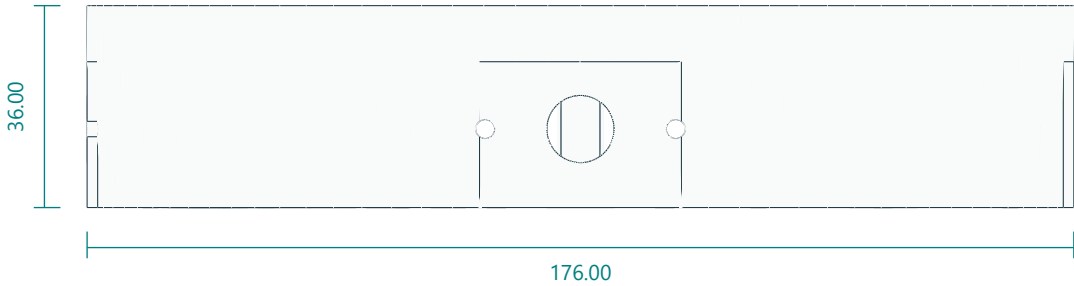
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

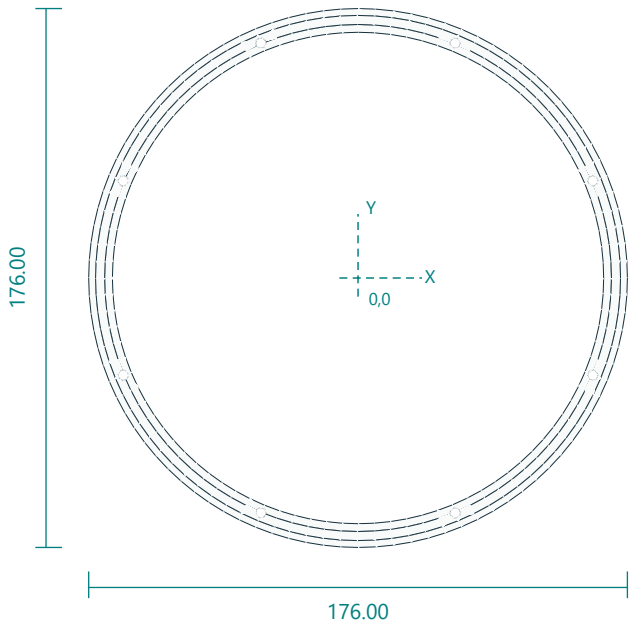


Interfaces: 176 diameter x36; rear thickness3; white carrier bosses radius45 top32.5.
Fasteners / retention: 8 M3x10 bezel; 4 M3x12 yoke.
Print orientation: rear disc down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Rear bolts(+/-27,+/-15); LCDpilots(+/-34,+/-34),top14. Whitecarrierpilotsradius45 cardinal,top32.5. Facebossradius60 angles45+90n,top35. Bezelradius83 angles22.5+45n,top35.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

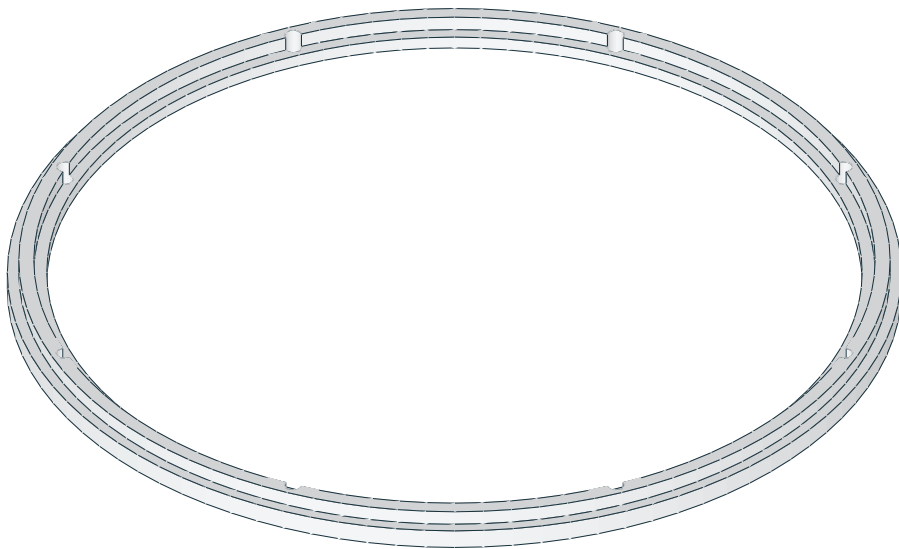
Outer RGB diffuser retaining bezel

outer_bezel.stl | PRINT QUANTITY 1 | PETG

TOP / XY



ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

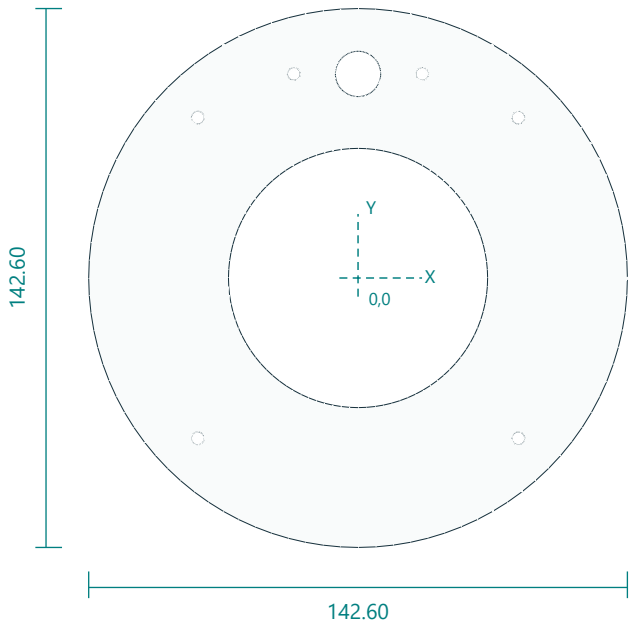


Interfaces: 176 diameter; window160.5; 8 holes radius83.
Fasteners / retention: 8 M3x10.
Print orientation: front lip down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Eight3.4 holes radius83 angles22.5+45n. Rearflangerecessdiameter165.5 by1.4 deep; facewindow160.5.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

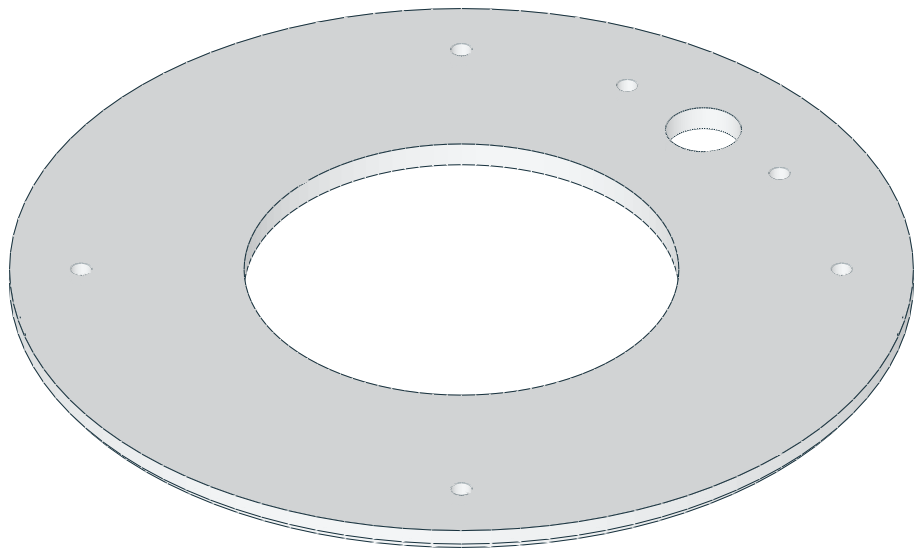
Central optical baffle plate

face_center.stl | PRINT QUANTITY 1 | black PETG

TOP / XY



ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

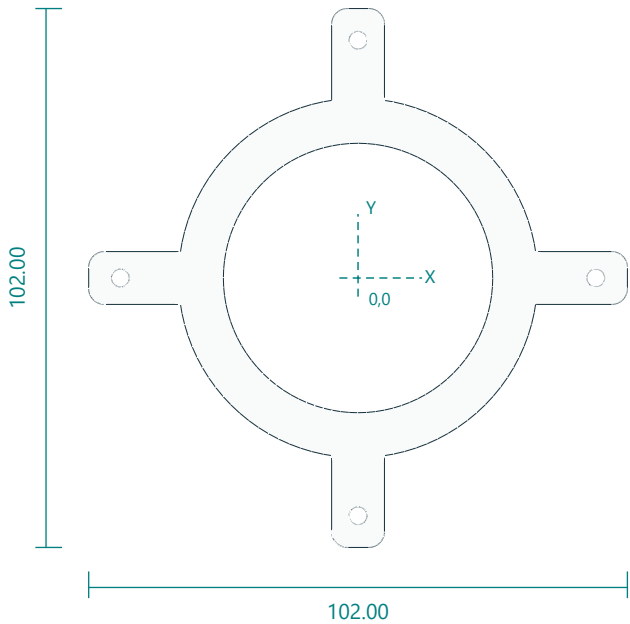


Interfaces: 142.6 diameter x4; center68.6; screws radius60.
Fasteners / retention: 4 M3x12 into shell bosses.
Print orientation: flat front down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holesradius60 angles45+90n. Frontpodholes(+/-17,54); cablehole(0,54)diameter12.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

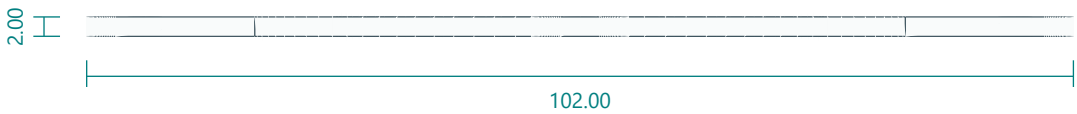
White ring carrier and mounting arms

white_carrier.stl | PRINT QUANTITY 1 | PETG

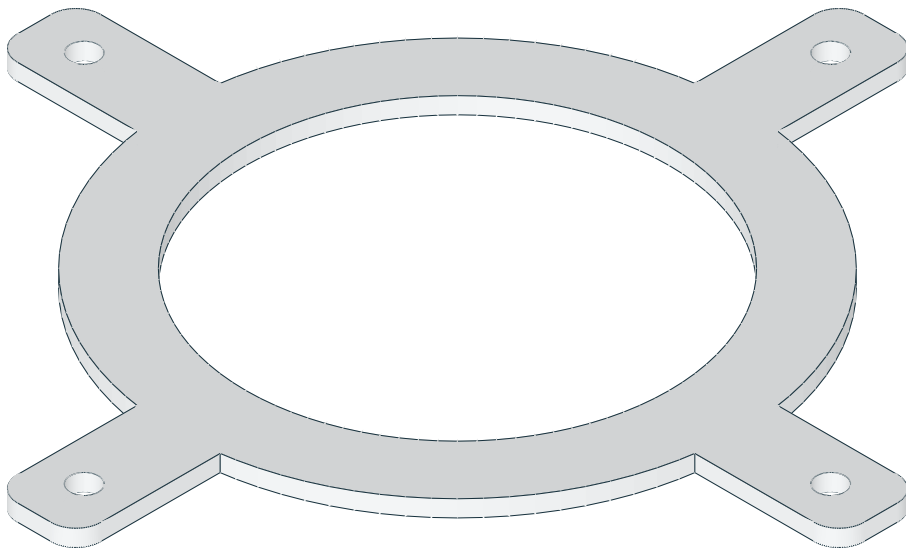
TOP / XY



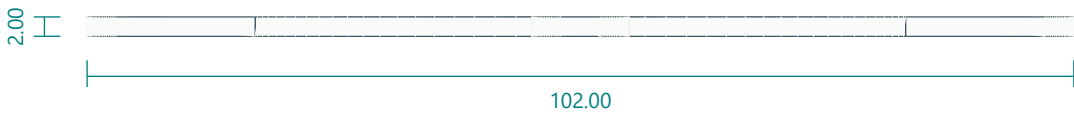
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

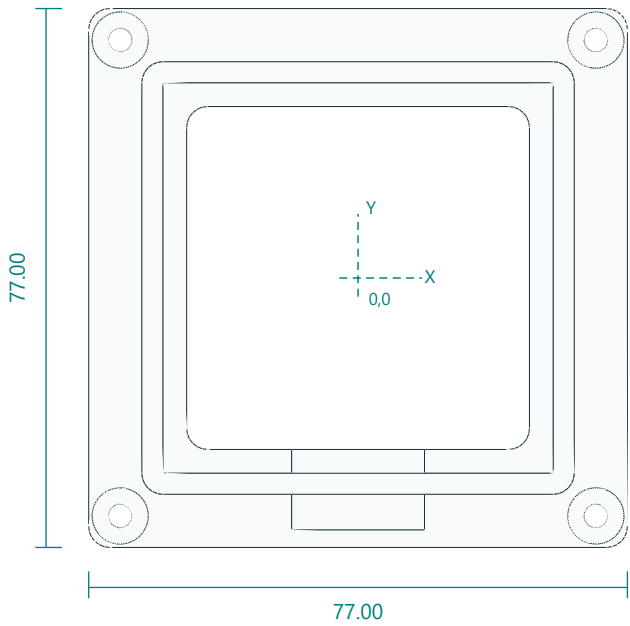


Interfaces: 68 OD51 ID; 4 cardinal mounting holes radius45; 2 thick.
Fasteners / retention: 4 M3x10.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes at(+/-45,0),(0,+/-45). Allarmsandring2mmthick.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

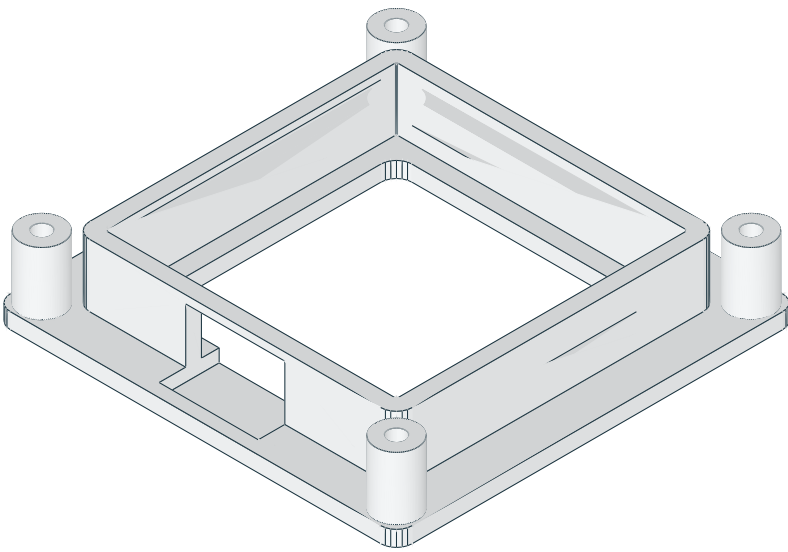
55 mm LCD edge cradle

lcd_cradle.stl | PRINT QUANTITY 1 | PETG

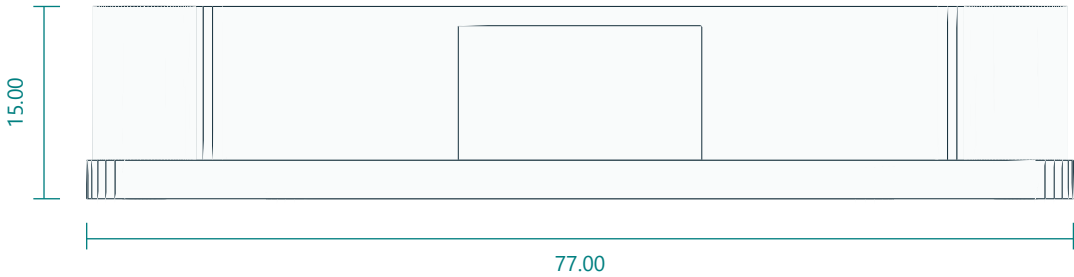
TOP / XY



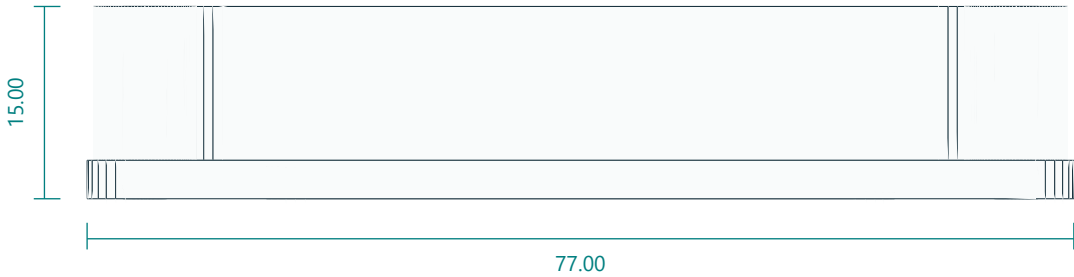
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

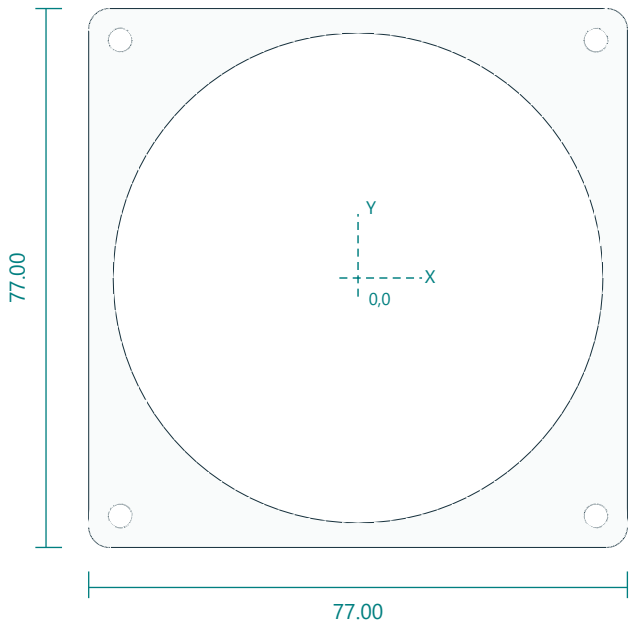


Interfaces: 55.8 square pocket; 15 depth; USB19 x13 relief.
Fasteners / retention: 4 M3x25; thin closed cell foam shims.
Print orientation: flat base down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes(+/-34,+/-34). Inner55.8square cavity. Rear49squareaccess. USBopening19wide at-Y.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

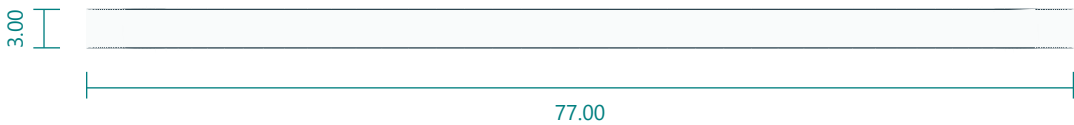
LCD PCB corner retaining plate

lcd_retainer.stl | PRINT QUANTITY 1 | black PETG

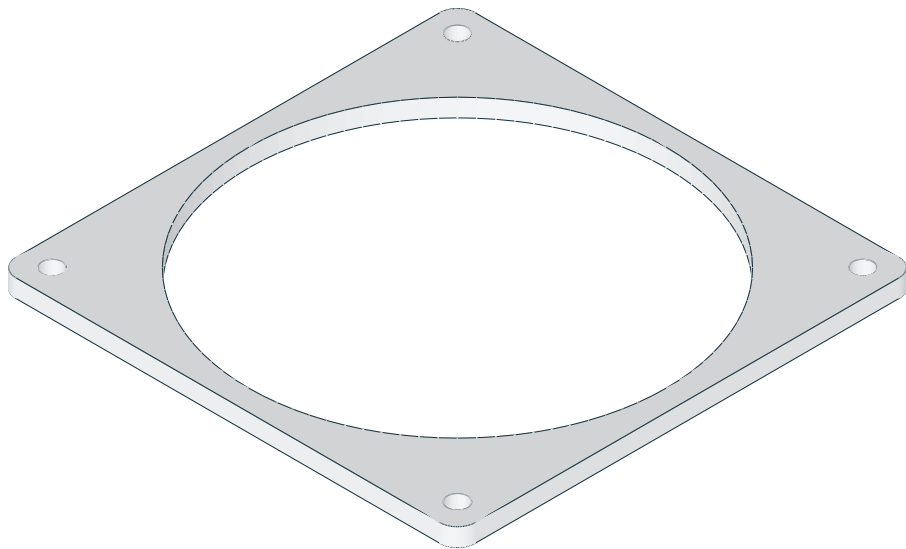
TOP / XY



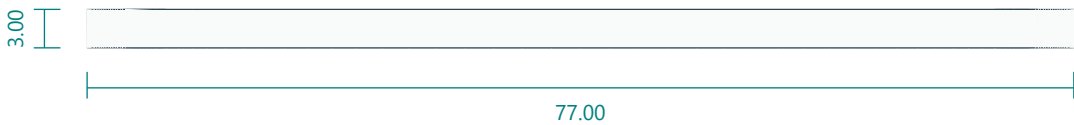
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

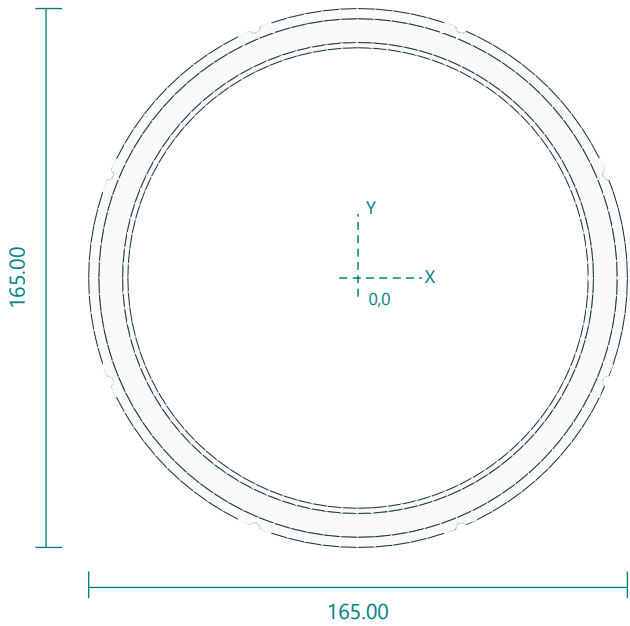


Interfaces: 77 square x3; opening70; mounting pitch68.
Fasteners / retention: 4 M3x25 shared.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Four3.4 holes(+/-34,+/-34). Diameter70centeropening retainsPCBcorners, not roundglass.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

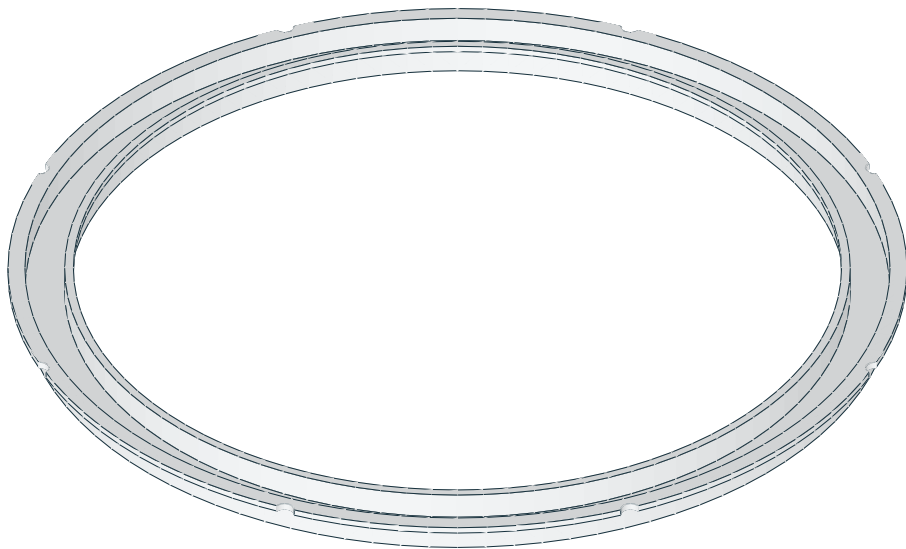
Frosted outer RGB annular diffuser

outer_diffuser.stl | PRINT QUANTITY 1 | natural/translucent PETG

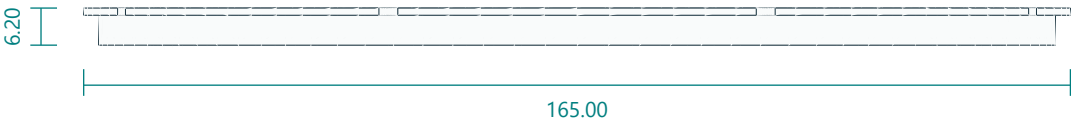
TOP / XY



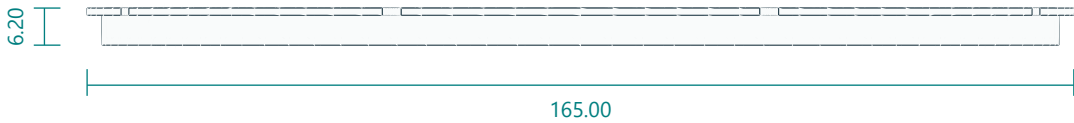
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

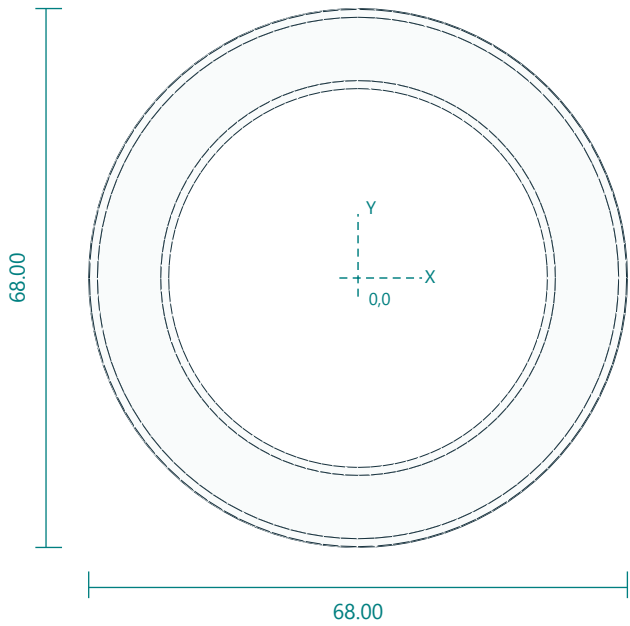


Interfaces: 160 face OD143 ID; 165 flange; 1.2 face; 6.2 depth.
Fasteners / retention: captured by outer bezel/center plate.
Print orientation: smooth optical face on bed. Layers 0.15 mm; 3 walls; 100% infill baseline.
Feature locations: FaceOD160 ID143. RearflangesOD165/ID158.8 andOD144.2/ID141. M3edgeclearancenotchesradius83.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

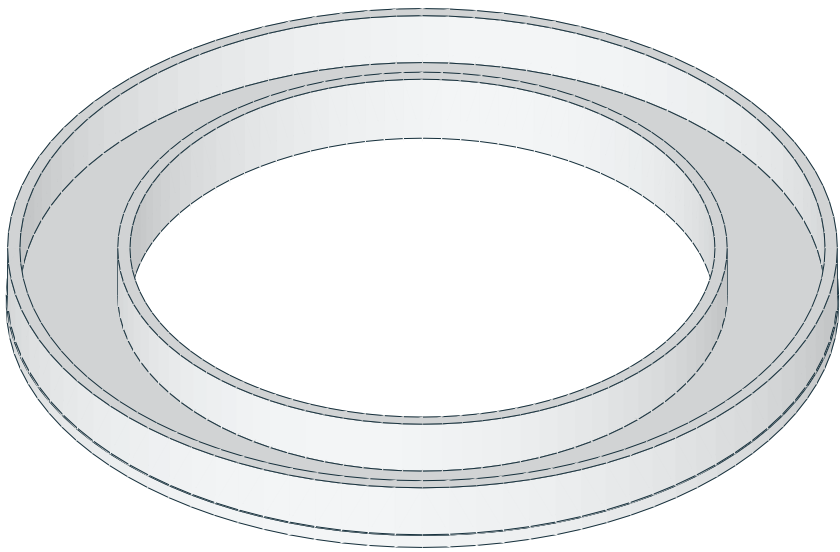
Frosted inner RGBW light diffuser

inner_diffuser.stl | PRINT QUANTITY 1 | natural/translucent PETG

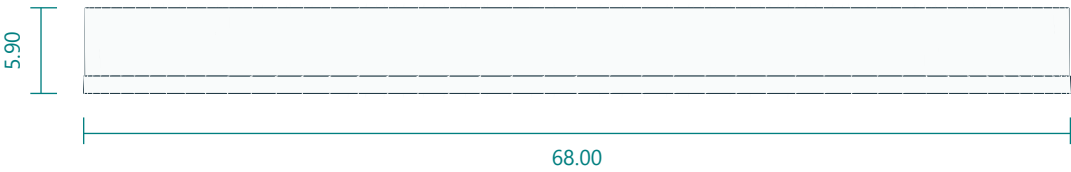
TOP / XY



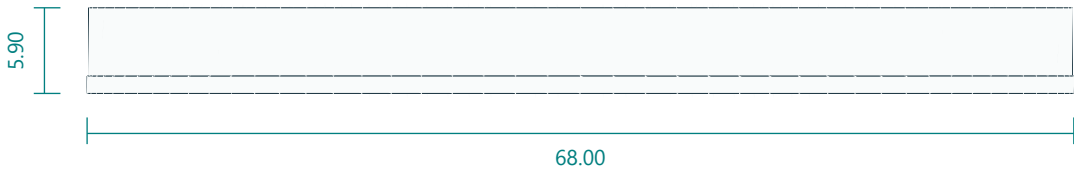
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

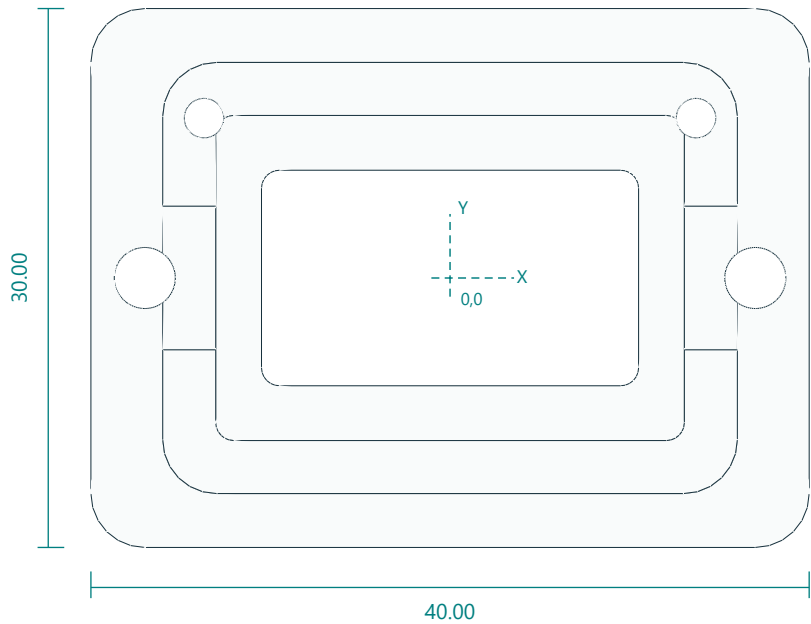


Interfaces: 68 OD47.8 ID; 1.2 face; 5.9 depth.
Fasteners / retention: 3 tiny neutral cure silicone retention dots.
Print orientation: smooth optical face on bed. Layers 0.15 mm; 3 walls; 100% infill baseline.
Feature locations: OD68 ID47.8; skin1.2; depth5.9. No sensorwindowmaterial.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

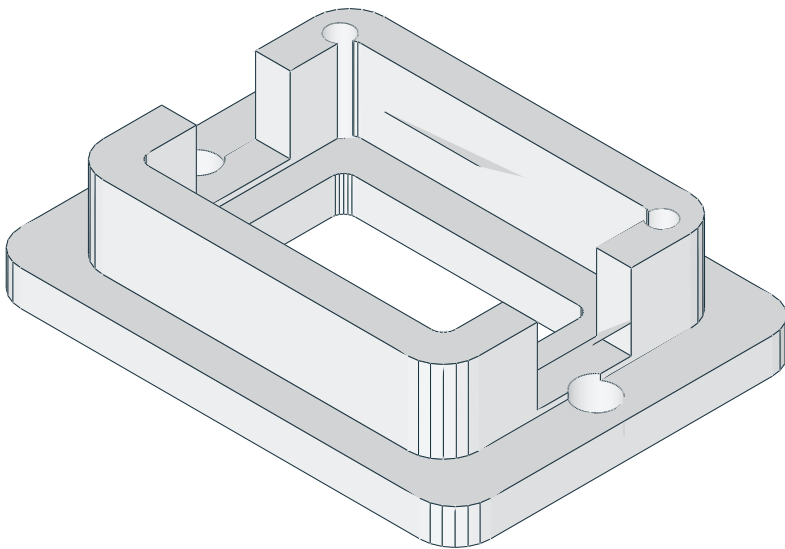
Open optical ToF sensor body

sensor_pod.stl | PRINT QUANTITY 5 | PETG

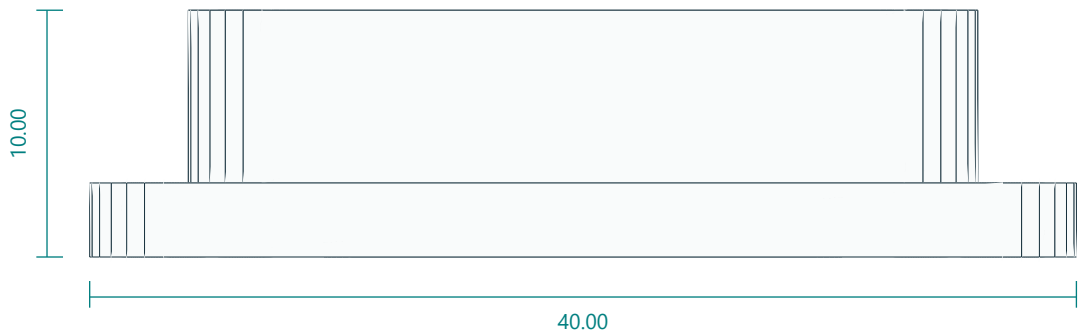
TOP / XY



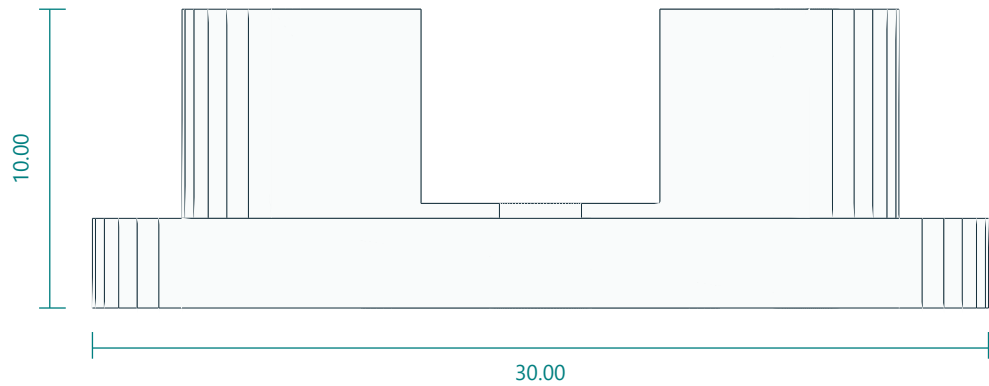
ISOMETRIC / REFERENCE



FRONT / XZ



RIGHT / YZ

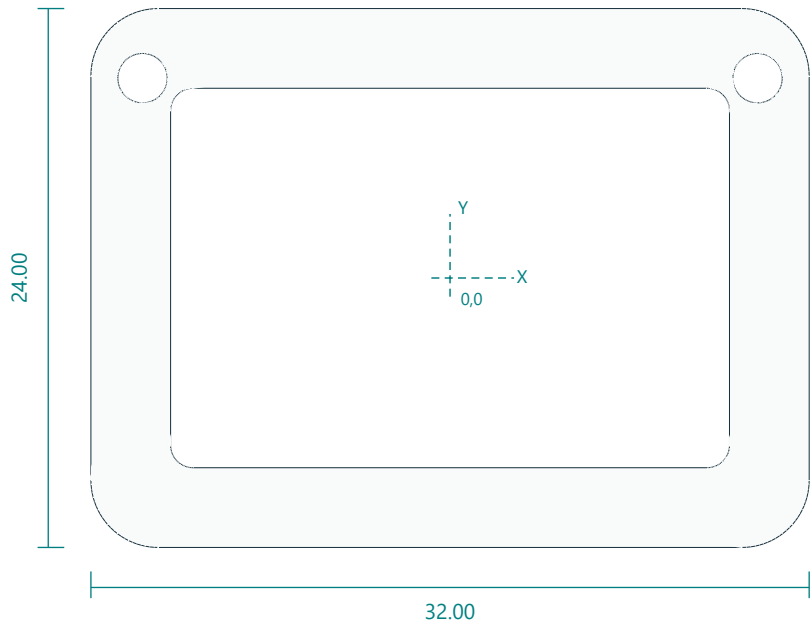


Interfaces: 40 x30 x10; PCB26.1 x18.1; mount pitch34; two side QT plug windows.
Fasteners / retention: 2 M3x10; 2 M2x14 each; two 18x2x6mm foam edge strips compressed to5.4mm.
Print orientation: rear flange down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Mount3.4 at(+/-17,0). M2holes2.2 at(+/-13.7,8.9). SideQTplugreliefs on +/-X, width8inY, Z3.5to10.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

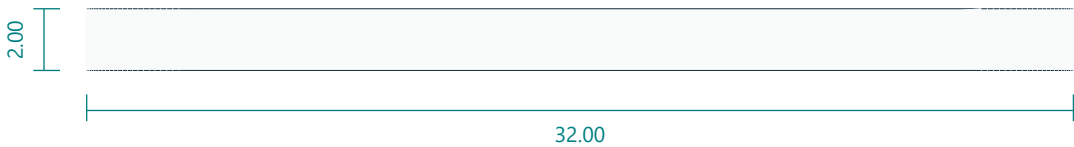
ToF board edge retaining frame

sensor_retainer.stl | PRINT QUANTITY 5 | PETG

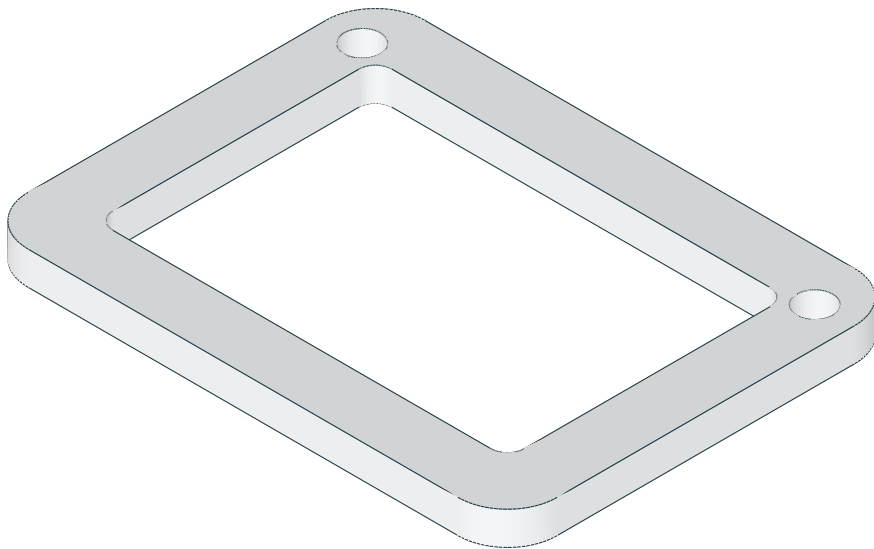
TOP / XY



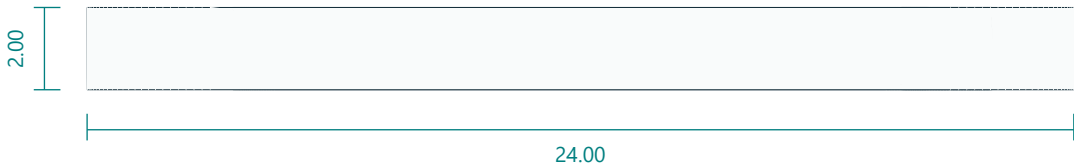
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

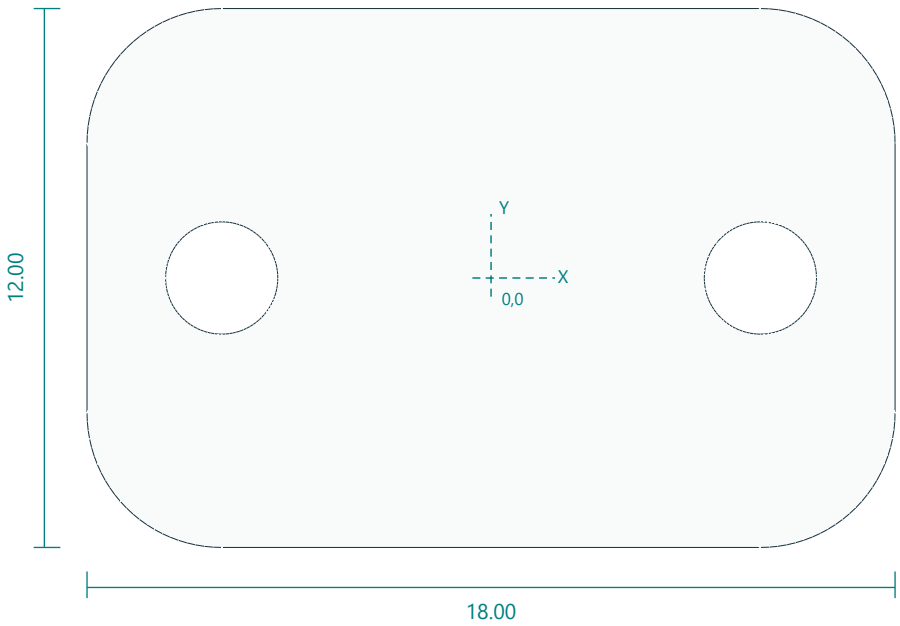


Interfaces: 32 x24 x2; unobscured aperture24.9 x16.9.
Fasteners / retention: 2 M2x14 each.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: M2holes2.2 at(+/-13.7,8.9). Opticalopening24.9x16.9.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

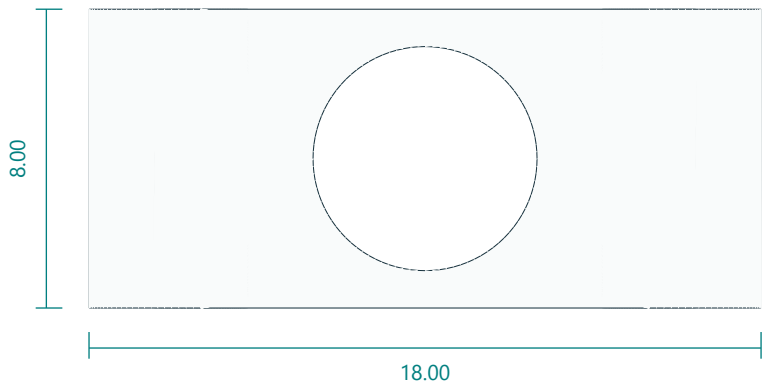
Harness saddle clip

cable_clip.stl | PRINT QUANTITY 6 | PETG

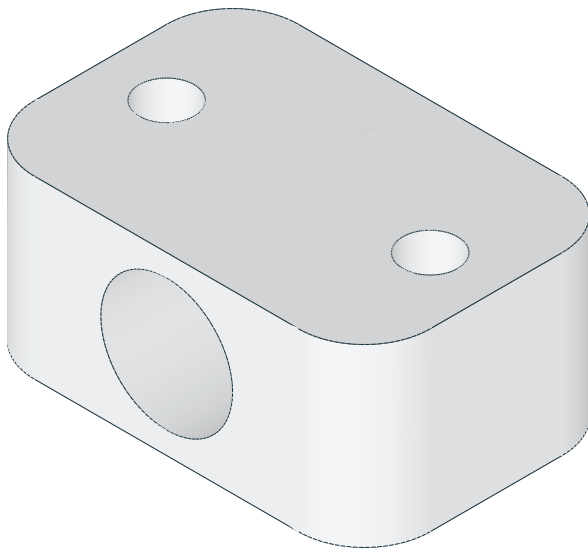
TOP / XY



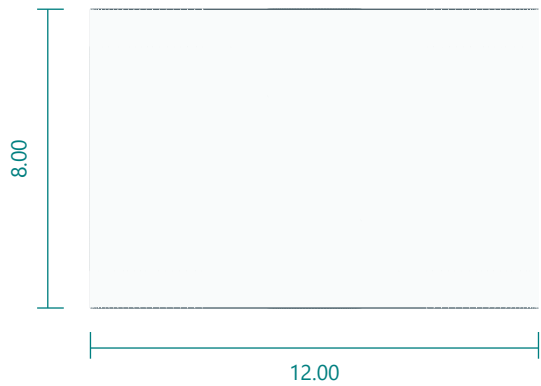
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ

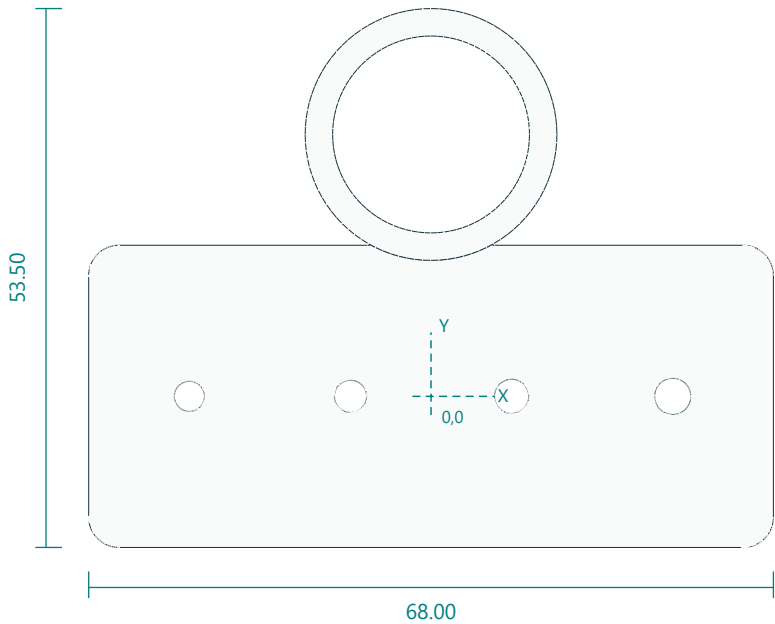


Interfaces: 18 x12 x8; cable6; screws12 pitch.
Fasteners / retention: 2 M2.5x12 each optional.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Two2.5holes(+/-6,0); cablebore6 alongY centeredZ4.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.

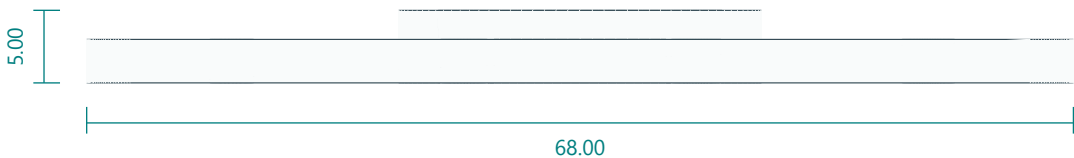
Horn and bolt clearance coupons

fit_coupon.stl | PRINT QUANTITY 1 | PETG

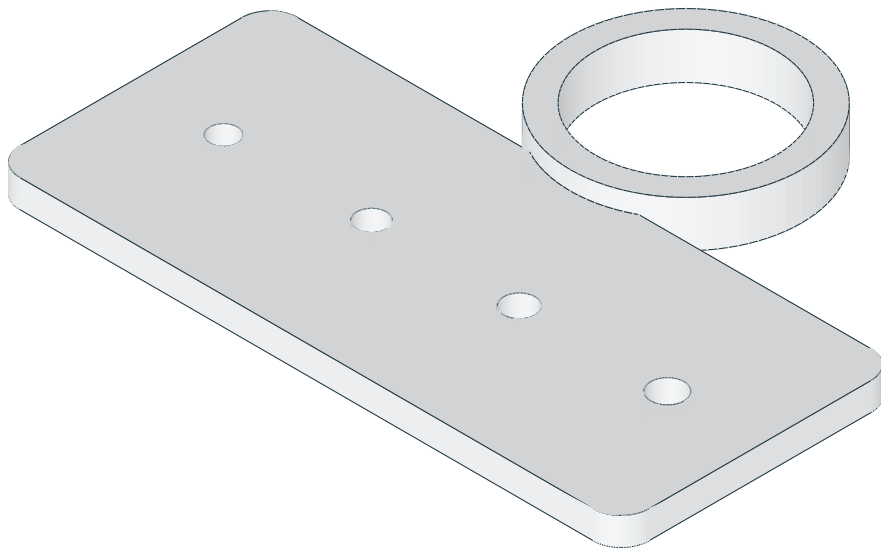
TOP / XY



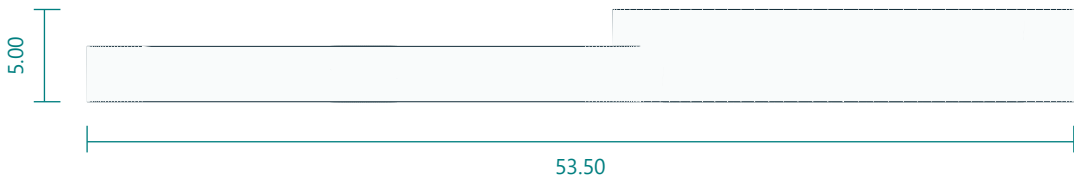
FRONT / XZ



ISOMETRIC / REFERENCE



RIGHT / YZ



Interfaces: 3.0 /3.2 /3.4 /3.6 bores; horn socket19.55.
Fasteners / retention: none.
Print orientation: flat down. Layers 0.2 mm; 4 walls; 25% infill baseline.
Feature locations: Bolt bores3.0,3.2,3.4,3.6 atX=-24,-8,8,24; hornsocket19.55 centeredY26.
General fit: inspect and deburr holes; check the fit coupon and actual hardware. No unlisted machining tolerances are implied.